Certified Speed-Limit Diagnostics for Deployed Superconducting Quantum Processors

Quantum speed limits · superconducting-qubit gate calibration

Carles Marín

Statements

Remark Abstract — Abstract 6
Remark §I — What the paper claims 7
Remark Table I — Table I 8
Theorem §II-A — Margolus–Levitin 10
Theorem §II-A — Mandelstam–Tamm, L^1 form 10
Theorem §II-A — Zieliński–Zych 11
Theorem §II-A — Two-level speed limit 12
Theorem §II-B — Entangling-time bound 13
Corollary §II-B — Bell preparation 14
Proposition §II-B — The bound is attained 15
Theorem §II-B — Chord-distance bound 15
Corollary §II-B — Consequence for CNOT and ECR 15
Theorem §II-B — Robust speed limits 16
Observation §III-C — The drive convention 17
Observation §III-C — The pulse-area identity 18
Observation §III-C — omegad is derived 18
Observation §III-D — Exclusions and falsifier 19
Observation §IV-A — The single-qubit gap 19
Observation §IV-B — Pre-registered hypotheses 21
Observation §IV-C — A cap near 0.1 21
Observation §IV-C — Per-qubit durations 22
Observation §V-A — The pairs 22
Observation §V-B — The two-qubit gap 22
Observation §V-C — The certified ratio 24
Observation §V-C — Numerical controls 24
Observation §V-D — Cross resonance and the first-order formula 24
Observation §VI — Median gaps by year 26
Observation §VI — Lengthening and error 27
Observation §VI — A methodological lesson 27
Observation §VII-A — The single-qubit law 28
Proposition §VII-A — Multilevel lower bound 30
Observation §VII-A — What IBM could reach 30
Observation §VII-A — Truncation 31
Observation §VII-A — The full amplitude of IBM 32
Observation §VII-B — Cross-resonance CNOT by optimal control 32
Observation §VII-B — CZ with the second level 33
Observation §VII-B — An explicit construction of the CZ 35
Observation §VII-B — The Choi-state bound for the gate 36
Observation §VII-B — The refined CZ bound 36
Observation §VII-B — Larger local dimension 38
Remark §VIII — The link and the model 39
Observation §VIII — Certified floor 40
Observation §VIII — Exact optimum 40
Remark Table VII — Table VII and the gaps 41
Remark §VIII — Caveats 42
Remark §IX — What the certificate guarantees 42
Remark §X — Reproducibility 43
Remark §XI — Conclusion 45
Remark §XI — Conflict of interest and use of AI assistants 46

At a glance

What the paper contributes

Two kinds of results, kept apart on this map. **Certified inequalities** (green, linked to Lean): the Margolus–Levitin bound, an L^1 form of Mandelstam–Tamm, the Zieliński–Zych family for every exponent $\alpha > 0$, time-dependent two-level bounds, and an entangling-time bound $JT \geq \arccos\lambda$ in arbitrary local dimension (a Bell state needs $JT \geq \pi/4$, attained; robust versions with state infidelity ε). **Diagnostics on deployed hardware** (grey, with the program that computes each number): how far the calibrated gates of IBM superconducting processors are from those bounds, from public, version-pinned calibration snapshots.

- Single-qubit X gates last a median 11.30 times their two-level speed limit $\pi/\Omega_{\max}$ (1646 qubits, 41 backends), a ratio frozen near 11 from 2021 to 2025.
- Two-qubit ECR/CX gates last a median 4.22 times the known CNOT-class bound $\pi/(2j_q)$; the ratio grew from 2.78 (2021) to 4.26 (2025) because gates got longer, not because couplings got weaker.
- Calibrated drives sit at fixed fractions of an energy splitting: $\Omega_{\text{peak}} \approx 0.09|\delta|$ (one qubit), $\Omega_{\text{CR}} \approx 0.45\Delta$ (two).
- The drive-strength convention is fixed by IBM's calibrated π pulses; the $\approx 15\%$ residual is explained by the pulse normalization: $\texttt{omegad} \cdot |\text{amp}| \cdot A_{\text{rec}} = \pi$ on all 777 qubits to $1.2 \cdot 10^{-7}$, and the residual tends to $1/(1-e^{-2}) = 1.15652$.
- At the drive IBM actually uses the X gate is within about 2 of the square-pulse limit (shape factor 1.87, or 2.16 with Qiskit's peak normalization); the 11.3 is against full drive amplitude, which IBM does not use.
- Optimal control on IBM's own transmon model finds feasible X gates of ≈ 16 ns at IBM's amplitude cap and 4.8 ns at its full amplitude (minimum at least 4.3 ns): 7.4–8.3 times faster than the archived 35.6 ns within the same amplitude budget; ≈ 3.4 ns needs three times that amplitude (numerical witnesses, not physical floors; the 6-level pulse is converged to 12 levels). The fit $\max(\pi/\Omega_{\max}, \sim 1.1 \cdot 2\pi/|\delta|)$ is empirical, not a theorem; $T \geq \pi/\Omega_{\max}$ (and $2\arcsin(1-e)/\Omega_{\max}$ in the trace metric) holds for any number of levels.
- With the second transmon level the minimal CZ time lies between 0.287 and $1/3$ of the qubit-only time: $0.4511/J \leq T_* \leq \pi/(6J)$ (refined bound, explicit construction; the certified norm bound is $\pi/(8J)$); optimal control reaches error $2 \cdot 10^{-8}$ at $0.55/J$ (0.35), and for the same task $0.4510/J \leq T_*(2 \cdot 10^{-8}) \leq 0.55/J$. In larger local dimension the construction and the entangling-time theorem approach each other (ratio $\to 1$, certified in Lean).

The paper claims no new physics: the speed limits are classical, the two-qubit constants are known for qubits, and the transmon model is IBM's. The entangling-time bound is, for qubits, a known corollary of Dür et al. and Vidal–Hammerer–Cirac, new only as a machine-checked, any-dimension assembly. This blueprint is built from a **draft**: the paper's own open items are shown where they matter.

🚀 **Start here:** **Observation §IV-A**: X gates last a median 11.30 times their two-level speed limit at full drive amplitude.

Main results:

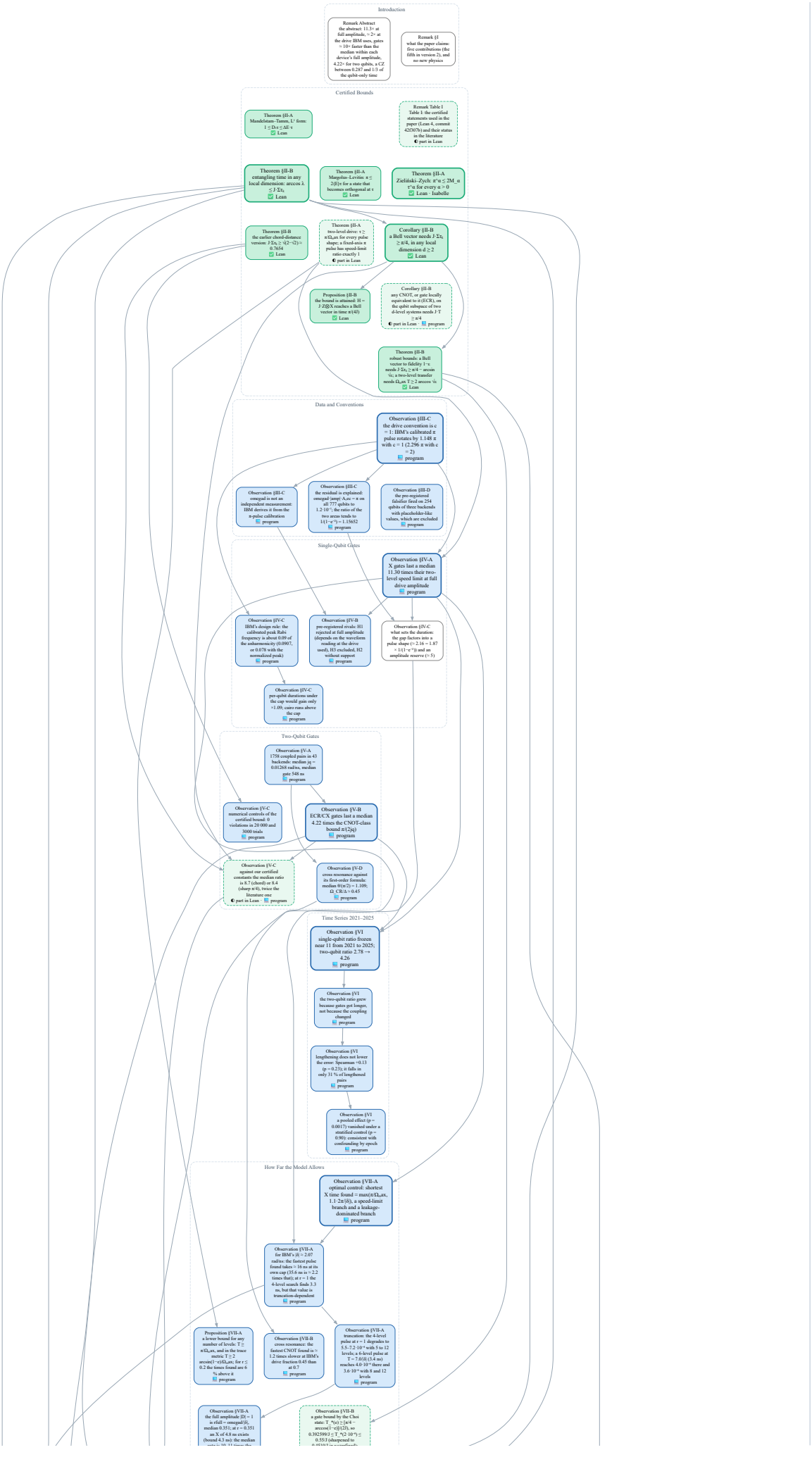
- **Theorem §II-A**: Zieliński–Zych: $\pi^{\wedge}\alpha \leq 2M_\alpha \tau^{\wedge}\alpha$ for every $\alpha > 0$
- **Theorem §II-B**: entangling time in any local dimension: $\arccos \lambda \leq J \cdot \Sigma\tau_k$
- **Corollary §II-B**: a Bell vector needs $J \cdot \Sigma\tau_k \geq \pi/4$, in any local dimension $d \geq 2$
- **Observation §III-C**: the drive convention is $c = 1$: IBM's calibrated π pulse rotates by 1.148 π with $c = 1$ (2.296 π with $c = 2$)
- **Observation §IV-A**: X gates last a median 11.30 times their two-level speed limit at full drive amplitude
- **Observation §V-B**: ECR/CX gates last a median 4.22 times the CNOT-class bound $\pi/(2jq)$
- **Observation §VI**: single-qubit ratio frozen near 11 from 2021 to 2025; two-qubit ratio 2.78 → 4.26
- **Observation §VII-A**: optimal control: shortest X time found $\approx \max(\pi/\Omega_{max}, 1.1 \cdot 2\pi/|\delta|)$, a speed-limit branch and a leakage-dominated branch
- **Observation §VII-B**: CZ with the second transmon level: error $2 \cdot 10^{-8}$ at $T = 0.55/J$ with local controls $\leq 10J$, i.e. 0.35 of the qubit-only time and 5 % above the construction $\pi/(6J)$

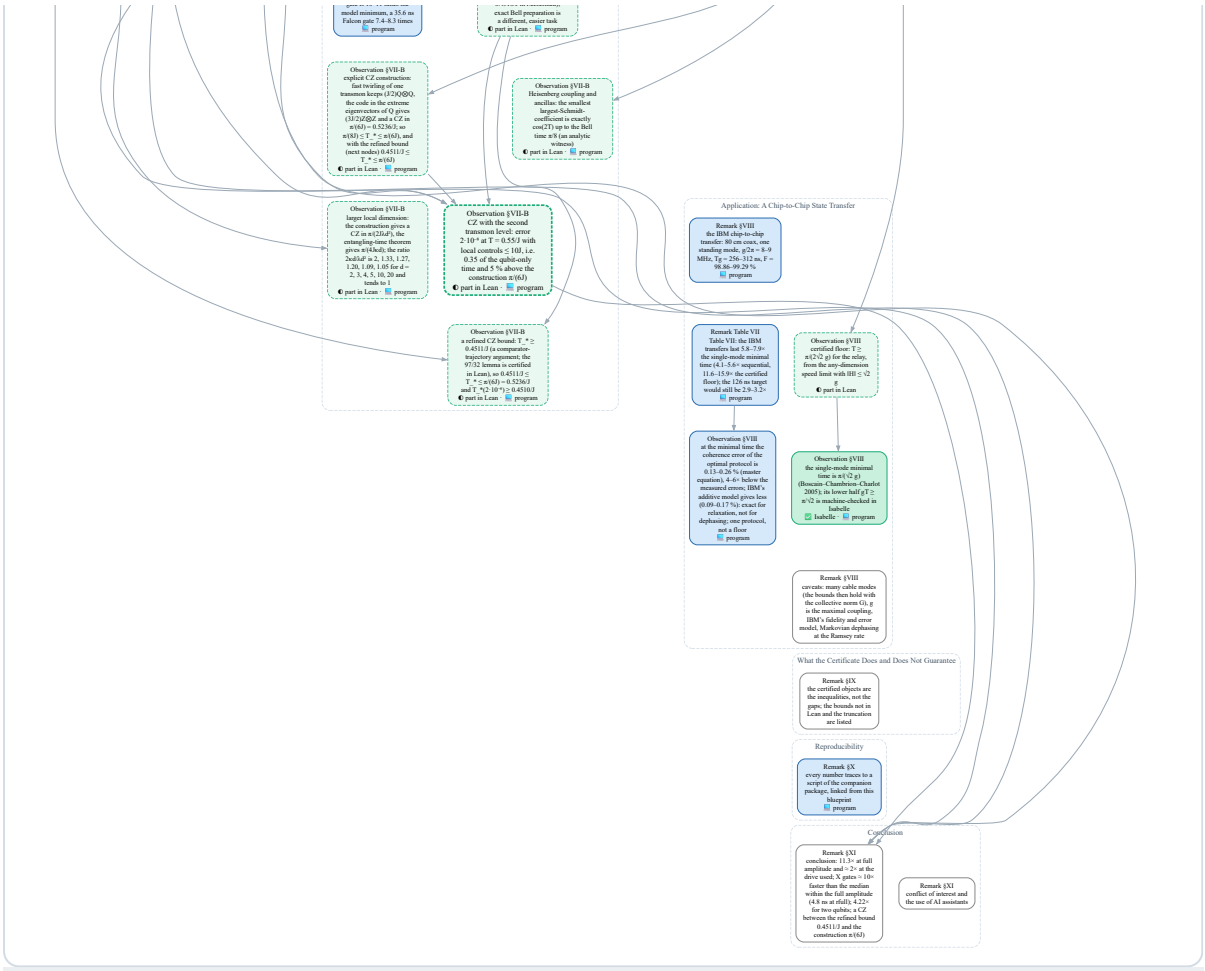
53 results | ✅ **8** entirely in Lean | ◐ **11** partly in Lean | ✅ **2** in Isabelle

💻 **35** checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.


Introduction
Remark Abstract
the abstract: 11.3× at full amplitude, ≈ 2× at the drive IBM uses, gates ≈ 10× faster than the median within each device's full amplitude, 4.22× for two qubits, a CZ between 0.287 and 1/3 of the qubit-only time
Remark §I
what the paper claims: five contributions (the fifth in version 2), and no new physics
Certified Bounds
Remark Table I
Table I: the certified statements used in the paper (Lean 4, commit 42f307b) and their status in the literature
◐ part in Lean
Theorem §II-A
Mandelstam–Tamm, L² form: 1 ≤ Dτ ≤ ΔE·τ
✅ Lean
Theorem §II-B
entangling time in any local dimension: arccos λ ≤ J·Σt₁
✅ Lean
Theorem §II-A
Margolus–Levitin: π ≤ 2⟨D⟩τ for a state that becomes orthogonal at τ
✅ Lean
Theorem §II-A
Zieliński–Zych: π^α ≤ 2M_α τ^α for every α > 0
✅ Lean · Isabelle
Theorem §II-B
the earlier chord-distance version: J·Σt₁ ≥ √(2−√2) ≈ 0.7654
✅ Lean
Theorem §II-A
two-level drive: τ ≥ π/Ω_max for every pulse shape; a fixed-area π pulse has speed-limit ratio exactly 1
◐ part in Lean
Corollary §II-B
a Bell vector needs J·Σt₀ ≥ π/4, in any local dimension d ≥ 2
✅ Lean
Proposition §II-B
the bound is attained: H = J·Z⊗X reaches a Bell vector in time π/(4J)
✅ Lean
Corollary §II-B
any CNOT, or gate locally equivalent to it (ECR), on the qubit subspace of two d-level systems needs J·T ≥ π/4
◐ part in Lean · program
Theorem §II-B
robust bounds: a Bell vector to fidelity 1−ε needs J·Σt₁ ≥ π/4 − arcsin √ε; a two-level transfer needs Ω_max·T ≥ 2 arccos √ε
✅ Lean
Data and Conventions
Observation §III-C
the drive convention is c = 1: IBM's calibrated π pulse rotates by 1.148 π with c = 1 (2.296 π with c = 2)
program
Observation §III-C
omegad is not an independent measurement: IBM derives it from the π-pulse calibration
program
Observation §III-C
the residual is explained: omegad/jump/Acos = π on all 777 qubits to 1.2·10⁻³; the ratio of the two areas tends to 1/(1−e⁻²) = 1.15652
program
Observation §III-D
the pre-registered falsifier fired on 254 qubits of three backends with placeholder-like values, which are excluded
program
Single-Qubit Gates
Observation §IV-A
X gates last a median 11.30 times their two-level speed limit at full drive amplitude
program
Observation §IV-C
IBM's design rule: the calibrated peak Rabi frequency is about 0.09 of the anharmonicity (0.0907, or 0.078 with the normalized peak)
program
Observation §IV-B
pre-registered rivals: H1 rejected at full amplitude (depends on the waveform reading at the drive used), H3 excluded, H2 without support
program
Observation §IV-C
what sets the duration: the gap factors into a pulse shape (≈ 2.16 = 1.87 × 1/(1−e⁻²)) and an amplitude reserve (≈ 5)
Observation §IV-C
per-qubit durations under the cap would gain only ≈1.09; cairo runs above the cap
program
Two-Qubit Gates
Observation §V-A
1758 coupled pairs in 43 backends: median jq = 0.01268 rad/ns, median gate 548 ns
program
Observation §V-C
numerical controls of the certified bound: 0 violations in 20 000 and 3000 trials
program
Observation §V-B
ECR/CX gates last a median 4.22 times the CNOT-class bound π/(2|q|)
program
Observation §V-C
against our certified constants the median ratio is 8.7 (chord) or 8.4 (sharp π/4), twice the literature one
◐ part in Lean · program
Observation §V-D
cross resonance against its first-order formula: median θ(π/2) = 1.109; Ω_CR/Δ = 0.45
program
Time Series 2021–2025
Observation §VI
single-qubit ratio frozen near 11 from 2021 to 2025; two-qubit ratio 2.78 → 4.26
program
Observation §VI
the two-qubit ratio grew because gates got longer, not because the coupling changed
program
Observation §VI
lengthening does not lower the error: Spearman +0.13 (p = 0.23); it falls in only 31 % of lengthened pairs
program
Observation §VI
a pooled effect (p = 0.0017) vanished under a stratified control (p = 0.96): consistent with confounding by epoch
program
How Far the Model Allows
Observation §VII-A
optimal control: shortest X time found = max(π/Ω_max, 1.1·2π/|δ|), a speed-limit branch and a leakage-dominated branch
program
Observation §VII-A
for IBM's |δ| = 2.07 rad/ns: the fastest pulse found takes ≈ 16 ns at its own cap (35.6 ns is = 2.2 times that); at r = 1 the 4-level search finds 3.3 ns, but that value is truncation-dependent
program
Proposition §VII-A
a lower bound for any number of levels: T ≥ π/Ω_max, and in the trace metric T ≥ 2 arccos(1−ε)/Ω_max; for r ≤ 0.2 the times found are 6 % above it
program
Observation §VII-B
cross resonance: the fastest CNOT found is ≈ 1.2 times slower at IBM's drive fraction 0.45 than at 0.7
program
Observation §VII-A
truncation: the 4-level pulse at r = 1 degrades to 5.5−7.2·10⁻⁴ with 5 to 12 levels; a 6-level pulse at T − 7.0/|δ| (3.4 ns) reaches 4.0·10⁻⁴ with 8 and 12 levels
program
Observation §VII-A
the full amplitude |Ω| = 1 is rfull = omegad/|δ|; median 0.351; at r = 0.351 an X of 4.8 ns exists (bound 4.3 ns); the median
Observation §VII-B
a gate bound by the Choi state: T_*(ε) ≥ [π/4 − arccos(1−ε)]/(2J), so 0.392590 ≤ T_*/2·10^3 ≤ 0.553 (sharpened to



📐 Setting (copied from the paper, §II and §III)

Units have $\hbar = 1$. (§II-A) Let a state have populations $p_n \geq 0$, $\sum_n p_n = 1$, on energies E_n, and autocorrelation $S(\tau) = \sum_n p_n e^{-iE_n\tau}$. If $S(\tau) = 0$ (orthogonality) at $\tau \geq 0$ the single-system bounds below apply. (§II-B) Two systems with finite index types (arbitrary local dimensions, ancillas included) evolve under $\prod_k \exp[-i\tau_k(A_k \otimes 1 + 1 \otimes B_k + H)]$, with A_k, B_k Hermitian and unbounded (piecewise-constant local control), $\tau_k \geq 0$, and a constant Hermitian interaction H with $\|H\| \leq J$ (operator norm). (§III-A) All data are public snapshots shipped with Qiskit under the Apache 2.0 licence: the fake backends of `qiskit-ibm-runtime` 0.51.0 (`props_*.json`: per-qubit T_1, T_2, frequency, anharmonicity, gate errors and durations; `conf_*.json`: the Duffing-model Hamiltonian variables w_q, δ, `omegad` and j_q in rad/ns), and the calibrated pulse definitions (`defs_*.json`, 32 backends) of `qiskit-terra` 0.24.1. (§III-B) IBM's own model is a Duffing oscillator per transmon, in the frame rotating at the qubit frequency with the rotating-wave approximation (RWA): $H = (\delta/2)\, n(n-1) + (\Omega/2)(b + b^\dagger)$, with $\Omega = c \cdot \texttt{omegad} \cdot |D|$ for a drive of complex amplitude D, and an exchange coupling $j_q(b_i^\dagger b_j + b_i b_j^\dagger)$ between coupled transmons. The speed limit of an X gate at full amplitude $|D| = 1$ is then $\tau_{\mathrm{QSL}} = \pi/(c \cdot \texttt{omegad})$ (and half of that for $\sqrt{X}$).

1 Introduction

Remark Abstract. (Abstract)

How far are deployed quantum gates from the fastest evolution quantum mechanics allows? We answer for pre-Heron IBM processors, mainly Falcon and Eagle, using public, version-pinned calibration snapshots and inequalities whose proofs are machine-checked in Lean 4. Heron snapshots lack a Hamiltonian. For single-qubit gates, with the drive-strength convention fixed by IBM's calibrated π pulses, the X gate lasts a median 11.3 times the two-level speed limit at full drive amplitude (1646 qubits, 41 backends). Most of this gap is a design choice: IBM drives at about a tenth of the anharmonicity, and at that drive its pulses are about 2 times the square-pulse limit. Within each device's full amplitude, optimal control on IBM's model gives gates about 10 times faster than the median. For two-qubit gates (1758 coupled pairs, 43 backends), the known qubit-subspace minimal time of controlled-NOT-class gates under exchange coupling (literature) gives a median ratio 4.22; across the snapshots it rose, not monotonically, from 2.78 (2021) to 4.26 (2025) as gates got longer, coinciding with the change from Falcon to Eagle. With the second level and fast arbitrary local control, the minimal time of a controlled-Z gate with ideal local control lies between 0.287 and $1/3$ of the qubit-only exchange time (refined bound, explicit construction); bounded local control reaches 0.35. We also formalize an entangling-time bound in arbitrary local dimension, known for qubits and new only as a machine-checked assembly. Applied to IBM's chip-to-chip state transfer (version 2), the same diagnostic puts the transfers at 5.8–7.9 times the single-mode minimal time. The contribution is a reproducible diagnostic built on the machine-checked inequalities, with explicit guarantees.

> 🆕 **Classical and new:** quoted from the current draft (27-oct-2026); every number in it is checked, node by node, below.

↳ Not used by another result of the paper.

Remark §I. (What the paper claims)

This paper contributes five things. *(1) Certified bounds.* The inequalities listed in Table I are proved in Lean 4 in the public development (the two-qubit gate bound $\pi/(2j_q)$ behind the headline 4.22 is taken from the literature and is not among them): Margolus–Levitin, an L^1 form of Mandelstam–Tamm, the Zieliński–Zych family for every exponent $\alpha>0$ (whose key auxiliary inequality the original paper states without proof), a time-dependent single-qubit bound with its equality case for fixed-axis pulses and its finite-tolerance version, an entangling-time bound in arbitrary local dimension with its shifted, locally equivalent (ECR) and finite-tolerance versions (Section II), and, for the CZ in d-level transmons, the norm bounds behind its asymptotic tightness and the key lemma of the refined bound (Section VII). *(2) A convention fixed by independent data.* The drive-strength parameter is pinned by IBM's calibrated π pulses through an exact pulse-area identity that holds on all 777 qubits with pulse definitions, up to which of two pulse normalizations reaches the qubit, which also reveals that it is derived from calibration rather than measured independently (Section III). *(3) A sweep of deployed hardware*: single-qubit gates on 1646 qubits, two-qubit gates on 1758 pairs, and a 2021–2025 time series (Sections IV–VI); for the single-qubit sweep, a falsifier and three rival hypotheses were written before the analysis, and we report their fate. *(4) Feasible times found by optimal control* on the vendor's own model, bracketed by lower bounds

(certified, or proved in the text) and, for the CZ, by an explicit construction, which in d-level transmons approaches the entangling-time bound as $d \to \infty$ at fixed interaction norm, with ideal local control (Section VII). *(5) An application beyond gates* (version 2): IBM's chip-to-chip state transfer runs at 5.8–7.9 times the single-mode minimal time, whose lower bound we machine-check in Isabelle; the master equation of the optimal protocol puts its coherence error four to six times below the measured errors (Section VIII). We claim no new physics: the speed limits are classical, the two-qubit constants are known for qubits, and the transmon model is IBM's. The exceptions are mathematical: the d-level quadrature code for the CZ and its asymptotic tightness were not found in the sources we read (Section VII). To our knowledge no prior work applies a proof-assistant-certified QSL to public calibration data with a per-gate gap; a search that finds nothing does not prove novelty. Section IX states what the certificate covers.

↳ Not used by another result of the paper.

2 Certified Bounds

Remark Table I. (Table I)

Certified statements used in this paper (Lean 4, commit `42f307b`; theorem names with prefix `QSL.`) and their status in the literature.

- `margolus_levitin`: $\pi \le 2\langle E\rangle\tau$. Status: classical (Margolus–Levitin).
- `mandelstam_tamm_L1`, `_weak`: $1 \le D_1\tau \le \Delta E\,\tau$. Status: weak form of Mandelstam–Tamm.
- `zielinski_zych_all`: $\pi^\alpha \le 2M_\alpha\tau^\alpha$, all $\alpha{>}0$. Status: stated in Zieliński–Zych; key inequality unproved there.
- `Qudit.arccos_le_entangling_time`: $\arccos\lambda \le J\sum\tau_k$. Status: corollary of Basyildiz–Jameson–Gong and Rudnicki.
- `Qudit.pi_div_four_le_entangling_time_fin`: Bell: $J\sum\tau_k \ge \pi/4$, any d . Status: $d = 2$: Dür et al., Vidal–Hammerer–Cirac.
- `Qudit.entangling_time_tight`: attained by $J\,Z\otimes X$. Status: $d = 2$: Dür et al., Vidal–Hammerer–Cirac.
- `Qudit.pi_div_four_le_entangling_time_local`, `_shift_local`: same for $(P\otimes Q)$ Bell (ECR). Status: corollary.
- `TwoQubit.entangling_time_lower_bound`: chord form, constant $\sqrt{2-\sqrt{2}}$. Status: weaker earlier version.
- `Qudit.arccos_le_entangling_time_shift`: $\|H\| \to \|H - c\mathbb{1}\|$. Status: corollary; $d = 2$: Dür et al.
- `Qudit.pi_div_eight_le_heisenberg_time`: Heisenberg Bell: $T \ge \pi/8$. Status: $d = 2$, no ancilla: Dür et al.
- `Gate.fsAngle_le_time_halfWidth`: angle $\le (\Omega_{\max}/2)\,T$, piecewise $H(t)$. Status: standard (shifted MT).

- `FixedAxis.mandelstam_tamm_integral_eq_solution`: $\int \Delta E = \arccos|\langle\psi_0|\psi_T\rangle|$, fixed axis. Status: equality case of Mandelstam–Tamm.
- `Qudit.pi_div_four_sub_le_entangling_time_approx`: Bell to fidelity $1-\varepsilon$: $J\sum\tau_k \geq \pi/4 - \arcsin\sqrt{\varepsilon}$. Status: robust corollary; formalized here.
- `Gate.two_arccos_sqrt_le_time_approx`: transfer to fidelity $1-\varepsilon$: $\Omega_{\max}T \geq 2\arccos\sqrt{\varepsilon}$. Status: two-level; tight.
- `Exchange.exchange_norm_ge`, `_le`, `ratio_tendsto_one`: $\lambda_d^2/2 \leq \kappa_d \leq 2(d-1)$; $2\kappa_d/\lambda_d^2 \to 1$. Status: not found in the sources read.
- `Exchange.refined_cz_lemma`: $\frac{1}{4}\mathrm{Tr}[(P_A\otimes P_B)H_c^2] \leq \frac{97}{32}J^2$, rank-2 codes, $d=3$. Status: lemma of the refined CZ bound.

Lean · ours, in part **show**

the statements of Table I (twenty Lean names), each checked by the Lean kernel in the public repository (commit 42f307b); the «status» column is the paper's reading of the literature, not a Lean statement

- `QSL.margolus_levitin`
- `QSL.mandelstam_tamm_L1`
- `QSL.mandelstam_tamm_weak`
- `QSL.zielinski_zych_all`
- `QSL.Qudit.arccos_le_entangling_time`
- `QSL.Qudit.pi_div_four_le_entangling_time_fin`
- `QSL.Qudit.entangling_time_tight`
- `QSL.Qudit.pi_div_four_le_entangling_time_local`
- `QSL.Qudit.pi_div_four_le_entangling_time_shift_local`
- `QSL.TwoQubit.entangling_time_lower_bound`
- `QSL.Qudit.arccos_le_entangling_time_shift`
- `QSL.Qudit.pi_div_eight_le_heisenberg_time`
- `QSL.Gate.fsAngle_le_time_halfWidth`
- `QSL.FixedAxis.mandelstam_tamm_integral_eq_solution`
- `QSL.Qudit.pi_div_four_sub_le_entangling_time_approx`
- `QSL.Gate.two_arccos_sqrt_le_time_approx`
- `QSL.Exchange.exchange_norm_ge`
- `QSL.Exchange.exchange_norm_le`
- `QSL.Exchange.ratio_tendsto_one`
- `QSL.Exchange.refined_cz_lemma`

✅ Checked when this page was built: ApproxSpeedLimits.lean, ExchangeQuadrature.lean, FixedAxisSaturation.lean, GateEigenphases.lean, MandelstamTamm.lean, MargolusLevitin.lean, QuditEntangling.lean, QuditEntanglingShift.lean, RefinedCZBound.lean, TwoQubitEntangling.lean, ZielinskiZych.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — margolus_levitin, mandelstam_tamm_L1, mandelstam_tamm_weak, zielinski_zych_all, arccos_le_entangling_time, pi_div_four_le_entangling_time_fin, entangling_time_tight, pi_div_four_le_entangling_time_local, pi_div_four_le_entangling_time_shift_local, entangling_time_lower_bound, arccos_le_entangling_time_shift, pi_div_eight_le_heisenberg_time, fsAngle_le_time_halfWidth, mandelstam_tamm_integral_eq_solution, pi_div_four_sub_le_entangling_time_approx, two_arccos_sqrt_le_time_approx, exchange_norm_ge, exchange_norm_le, ratio_tendsto_one, refined_cz_lemma: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean`

`ApproxSpeedLimits.lean` `ExchangeQuadrature.lean` `FixedAxisSaturation.lean` `GateEigenphases.lean` `MandelstamTamm.lean` `MargolusLevitin.lean` `QuditEntangling.lean` `QuditEntanglingShift.lean` `RefinedCZBound.lean` `TwoQubitEntangling.lean` `ZielinskiZych.lean` (SHA-256 of the compiled file: `224c15dcafc7d45b…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** status in the literature as printed in Table I: classical, weak form of Mandelstam–Tamm, «stated in [ZZ]; key inequality unproved there», corollary of Basyildiz–Jameson–Gong and Rudnicki, d = 2 known (Dür et al.; Vidal–Hammerer–Cirac), standard (shifted MT), equality case of MT, «robust corollary; formalized here» for the two tolerance (ε) theorems, «not found in the sources read» for the exchange-norm theorems, and «lemma of the refined CZ bound» for the 97/32 lemma.

↳ Not used by another result of the paper.

Theorem §II-A. ✅ Lean **(Margolus–Levitin)✓**

> *Let a state have populations $p_n \geq 0$, $\sum_n p_n = 1$, on energies E_n, and autocorrelation $S(\tau) = \sum_n p_n e^{-iE_n\tau}$. If $S(\tau) = 0$ (orthogonality) at $\tau \geq 0$ and $E_n \geq 0$, then $\pi \leq 2\langle E\rangle\tau$.*

Lean · ours show

- `QSL.margolus_levitin`

✅ Checked when this page was built: MargolusLevitin.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — margolus_levitin: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean MargolusLevitin.lean` (SHA-256 of the compiled file: `224c15dcafc7d45b…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** classical (Margolus and Levitin, 1998); Table I: «classical».

↳ Used in: Theorem §II-A (p. 12)

Proof

> Formalized in Lean (see the linked declarations).

Theorem §II-A. ✅ Lean **(Mandelstam–Tamm, L¹ form)✓**

> *If $S(\tau) = 0$ at $\tau \geq 0$, then $1 \leq D_1\tau$ with $D_1 = \sum_n p_n|E_n - \bar{E}|$; and $1 \leq \Delta E\,\tau$. The full MT constant $\tau \geq \pi/(2\Delta E)$ is not proved in the development.*

Lean · ours show

- `QSL.mandelstam_tamm_L1`
- `QSL.mandelstam_tamm_weak`

✅ Checked when this page was built: MandelstamTamm.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — mandelstam_tamm_L1, mandelstam_tamm_weak: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean`

MandelstamTamm.lean (SHA-256 of the compiled file: 457f13a1a2df01b9…; a source page on this site may show it with comments edited, under its own hash).

> NEW **Classical and new:** a weak form of Mandelstam–Tamm (Table I). The paper: the full constant $\tau \geq \pi/(2\Delta E)$ is not proved in the development.

↳ Not used by another result of the paper.

Proof

> Formalized in Lean (see the linked declarations).

Theorem §II-A. ✅ Lean · Isabelle **(Zieliński–Zych)✓**

> *If $S(\tau) = 0$ at $\tau \geq 0$, with $E_n \geq E_0$ on populated levels and $M_\alpha = \sum_n p_n (E_n - E_0)^\alpha$, then $\pi^\alpha \leq 2 M_\alpha \tau^\alpha$ for every $\alpha > 0$. The auxiliary inequality $x^\alpha - \pi^\alpha/2 + (\pi^\alpha/2)\cos x + \alpha\pi^{\alpha-1}\sin x \geq 0$ $(x \geq 0)$ is stated without proof in the original paper for all $\alpha > 0$; Lean proves it for $0 < \alpha \leq 1$ and for $\alpha \geq 1$.*

Lean · ours **show**

- QSL.zielinski_zych_all
- QSL.zielinski_zych_time_all
- QSL.zz_inequality
- QSL.zz_inequality_of_one_le

Isabelle · ours **show**

zielinski_zych · Isabelle/AFP/Quantum_Speed_Limits/Zielinski_Zych.thy, line 464 — the bound for 0 < α ≤ 1 (entry Quantum_Speed_Limits submitted to the AFP, 30-oct-2026); the Lean development covers every α > 0

```
theorem zielinski_zych:
  fixes p E :: "nat \<Rightarrow> real" and e0 a M ta :: real
  assumes a0: "0 < a" and a1: "a \<le> 1"
    and hp: "\<forall>n. 0 \<le> p n" and hE: "\<forall>n. e0 \<le> E n" and
hta: "0 \<le> ta"
    and hsum: "p sums 1"
    and hM: "(\<lambda>n. p n * (E n - e0) powr a) sums M"
    and hRe: "(\<lambda>n. p n * cos (E n * ta)) sums 0"
    and hIm: "(\<lambda>n. p n * sin (E n * ta)) sums 0"
  shows "pi powr a \<le> 2 * M * ta powr a"
```

✅ Checked when this page was built: ZielinskiZych.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — zielinski_zych_all, zielinski_zych_time_all, zz_inequality, zz_inequality_of_one_le: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean ZielinskiZych.lean` (SHA-256 of the compiled file: 413bdc0e08aba2ad…; a source page on this site may show it with comments edited, under its own hash).

> NEW **Classical and new:** stated in Zieliński and Zych (2006); the key inequality is stated there without proof for all α > 0 (Table I); Lean proves it for 0 < α ≤ 1 and for α ≥ 1.

↳ Not used by another result of the paper.

Proof

> Formalized in Lean (see the linked declarations).

In plain words (commentary, not part of the paper)

What it says. A quantum state cannot turn into a state orthogonal to it faster than its energy allows. If the energy above the ground level E_0 is spread as p_n over levels E_n, the time τ to reach an orthogonal state obeys $\pi^\alpha \le 2M_\alpha \tau^\alpha$, where M_α is the α-th moment of that energy. For $\alpha = 1$ this is the Margolus–Levitin bound; the statement here holds for *every* exponent $\alpha>0$.

Why it is true. The paper puts the weight on one auxiliary inequality, $x^\alpha - \pi^\alpha/2 + (\pi^\alpha/2)\cos x + \alpha\pi^{\alpha-1}\sin x \ge 0$ for $x \ge 0$, and says that the original paper states it without proof for all $\alpha>0$, while the Lean development proves it in two ranges, $0<\alpha \le 1$ and $\alpha \ge 1$. (Supposed, not written in the paper: the bound follows by putting $x = (E_n - E_0)\tau$ and averaging over p_n.)

What is new and what is not. Table I of the paper lists the statement as "stated in [ZZ]; key inequality unproved there": the claim of the paper is the proof, checked by the Lean kernel, not the inequality. **Numbers.** None: this is a theorem, not a measurement.

Theorem §II-A. (Two-level speed limit)

> *For a two-level system driven at constant Rabi frequency Ω, ML gives $\tau \ge \pi/\Omega$ for $|0\rangle \to |1\rangle$, saturated by a square π pulse. For a time-dependent drive two further statements are certified. (i) Any piecewise-constant evolution whose Hamiltonians satisfy $\|H_k - c_k\mathbb{1}\| \le \Omega_{\max}/2$ for some real c_k moves any state by a Fubini–Study angle at most $(\Omega_{\max}/2)\,T$; hence $|0\rangle \to |1\rangle$ needs $T \ge \pi/\Omega_{\max}$ for every pulse shape. (ii) For a fixed-axis drive $H(t) = f(t)\,\sigma_x/2$ with $f \ge 0$ and $\int f \le \pi$, starting from $|0\rangle$, the generalized MT bound $\int \Delta E\, dt \ge \arccos|\langle\psi_0|\psi_T\rangle|$ holds with equality: every fixed-axis π pulse has speed-limit ratio exactly 1, whatever its shape. It does not extend to DRAG pulses.*

Lean · ours, in part **show**

constant Rabi frequency: Margolus–Levitin. Time-dependent drive: (i) piecewise-constant evolutions with spectral width at most Ω_max move any state by a Fubini–Study angle at most (Ω_max/2)T; (ii) fixed-axis drive: equality case of the generalized Mandelstam–Tamm bound; (iii) transfer to fidelity 1−ε: Ω_max T ≥ 2 arccos √ε. Continuous pulses are covered only as limits of piecewise-constant ones (not formalized).

- `QSL.margolus_levitin`
- `QSL.Gate.fsAngle_le_time_halfWidth`
- `QSL.FixedAxis.mandelstam_tamm_integral_eq_solution`
- `QSL.Gate.two_arccos_sqrt_le_time_approx`

✅ Checked when this page was built: ApproxSpeedLimits.lean, FixedAxisSaturation.lean, GateEigenphases.lean, MargolusLevitin.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — margolus_levitin, fsAngle_le_time_halfWidth, mandelstam_tamm_integral_eq_solution, two_arccos_sqrt_le_time_approx: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean ApproxSpeedLimits.lean FixedAxisSaturation.lean GateEigenphases.lean`

`MargolusLevitin.lean` (SHA-256 of the compiled file: `224c15dcafc7d45b…`; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** the single-qubit reference time used in Section IV; (i) is the standard shifted Mandelstam–Tamm bound, (ii) the equality case of Mandelstam–Tamm, behind the near-unit ratios of Ishida–Hasegawa for π pulses; it does not extend to DRAG pulses.

↳ Used in: §IV-A (p. 19), Proposition §VII-A (p. 30)

Theorem §II-B. ✅ Lean **(Entangling-time bound)** ✓

> *Two systems with finite index types (arbitrary local dimensions, ancillas included) evolve under* $\prod_k \exp[-i\tau_k(A_k \otimes 1 + 1 \otimes B_k + H)]$*, with* A_k, B_k *Hermitian and unbounded (piecewise-constant local control),* $\tau_k \geq 0$*, and a constant Hermitian interaction* H *with* $\|H\| \leq J$ *(operator norm). If the evolution maps a unit product vector to* Φ*, and every unit product vector has overlap at most* λ *with* Φ*, then* $\arccos \lambda \leq J \sum_k \tau_k$.

Lean · ours **show**

- `QSL.Qudit.arccos_le_entangling_time`
- `QSL.Qudit.arccos_le_entangling_time_shift`

✅ Checked when this page was built: QuditEntangling.lean, QuditEntanglingShift.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — arccos_le_entangling_time, arccos_le_entangling_time_shift: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean QuditEntangling.lean QuditEntanglingShift.lean` (SHA-256 of the compiled file: `f4a5538fe9427bac…`; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** a corollary of Basyildiz–Jameson–Gong and Rudnicki (Table I); «new only as a machine-checked, any-dimension assembly»; for qubits a known corollary of Dür et al. and Vidal–Hammerer–Cirac.

↳ Used in: Corollary §II-B (p. 14), §VII-B (p. 33), §VII-B (p. 35), §VII-B (p. 36), §VII-B (p. 38), §VII-B

Proof

> Formalized in Lean (see the linked declarations).

💡 **In plain words (commentary, not part of the paper)**

What it says. Take two quantum systems of any size, each with as many extra "ancilla" levels as you like, and let them interact through a fixed coupling H whose strength is at most J. You may act on each system separately as violently and as fast as you want. Start from an unentangled (product) state. If the state you want to reach is so entangled that every unentangled state overlaps it by at most λ, then the total interaction time $\sum_k \tau_k$ is at least $\arccos(\lambda)/J$. Local control is free; only the coupling costs time.

Why it is true. The paper names the method: a Duhamel estimate in the Fubini–Study angle, chained over the control steps. (Supposed, not written in the paper: local steps leave the angle to the set of product states unchanged, and the interaction moves it at rate at most J.)

What is new and what is not. The paper says this is, for qubits, a known corollary of Dür–Vidal–Cirac–Linden– Popescu and Vidal–Hammerer–Cirac, and of the Basyildiz–Jameson–Gong machinery and Rudnicki's Schmidt-plus-arccos step. What it adds is the machine-checked statement for arbitrary local dimension and ancillas. It also says the initial state must be a product: with pre-entangled ancillas the argument does not apply.

Numbers. None here. The bound is exact for the Bell state (next node).

Corollary §II-B. ✅ Lean **(Bell preparation)✓**

> *The overlap of a state with product vectors is bounded by its largest Schmidt coefficient. For a two-level Bell vector $(e_{00}+e_{11})/\sqrt{2}$ inside qudits of any dimension $d \geq 2$, $\lambda = 1/\sqrt{2}$, hence $J\sum_k \tau_k \geq \pi/4$.*

Lean · ours **show**

- `QSL.Qudit.pi_div_four_le_entangling_time_fin`

✅ Checked when this page was built: QuditEntangling.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — pi_div_four_le_entangling_time_fin: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean QuditEntangling.lean` (SHA-256 of the compiled file: `f4a5538fe9427bac…`; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** for d = 2 known with equality: Dür et al. (2001) and Vidal–Hammerer–Cirac (2002) (Table I).

↳ Used in: Proposition §II-B (p. 15), Corollary §II-B (p. 15), Theorem §II-B (p. 16), §V-C (p. 24)

Proof

> Formalized in Lean (see the linked declarations).

💡 **In plain words (commentary, not part of the paper)**

What it says. Creating a maximally entangled pair (a Bell state) from an unentangled one costs interaction time at least $\pi/(4J)$, in whatever dimension the two systems live. The reason is that the Bell state overlaps every product state by at most $1/\sqrt{2}$, and $\arccos(1/\sqrt{2}) = \pi/4$. The next node shows the bound is attained.

Why it is true. It is the previous theorem with $\lambda = 1/\sqrt{2}$; the paper gives that value for a two-level Bell vector inside qudits of any dimension $d \geq 2$ (the overlap of a state with product vectors is bounded by its largest Schmidt coefficient).

What is new and what is not. For $d = 2$ the result is known with equality (Dür et al., Vidal–Hammerer–Cirac); the paper's own words: "new only as a machine-checked, any-dimension assembly".

A consequence the paper draws (and its limit). CNOT maps $|+\rangle|0\rangle$ to a Bell state, so any CNOT on the qubit subspace of two d-level systems needs $JT \geq \pi/4$, with J the norm of the interaction on the full d-level space. A numerical check (`cnot_m3.py`) gives the overlap constants $m_2 = 0.707107$ and $m_3 = 0.707105$ over 45 starts, i.e. $T \geq 0.785401/J$, matching $1/\sqrt{2}$ (numbers quoted from the paper; a re-run for this blueprint reproduced $m_2 = 0.707107$ but its m_3 search did not finish within ten minutes). The CNOT corollary

itself is the Bell bound applied to $|+\rangle|0\rangle$; the extension to gates only locally equivalent to CNOT, such as ECR, is a Lean theorem (`pi_div_four_le_entangling_time_local`).

Proposition §II-B. Lean **(The bound is attained)**✓

> *The bound is attained: $H = J\,Z \otimes X$ reaches the Bell vector from a product vector in one step of length $\pi/(4J)$.*

Lean · ours **show**

- `QSL.Qudit.entangling_time_tight`

✅ Checked when this page was built: QuditEntangling.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — entangling_time_tight: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean QuditEntangling.lean` (SHA-256 of the compiled file: `f4a5538fe9427bac…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** for d = 2: Dür et al. and Vidal–Hammerer–Cirac (Table I).

↳ Not used by another result of the paper.

Proof

Formalized in Lean (see the linked declarations).

Theorem §II-B. Lean **(Chord-distance bound)**✓

> *An earlier chord-distance version in the same development gives the weaker constant $\sqrt{2-\sqrt{2}} \approx 0.7654$, 2.6% below the exchange-coupling time $\pi/(4J) = 0.7854/J$.*

Lean · ours **show**

- `QSL.TwoQubit.entangling_time_lower_bound`

✅ Checked when this page was built: TwoQubitEntangling.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — entangling_time_lower_bound: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean TwoQubitEntangling.lean` (SHA-256 of the compiled file: `d1b9ef08f2a9b3aa…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** superseded by the sharp constant π/4 in the same development; 2.6 % below the exchange-coupling time $\pi/(4J) = 0.7854/J$.

↳ Used in: §V-C (p. 24), §V-C (p. 24)

Proof

Formalized in Lean (see the linked declarations).

Corollary §II-B. (Consequence for CNOT and ECR)

> *CNOT maps $|+\rangle|0\rangle$ to the Bell state, so any implementation of CNOT on the qubit subspace of two d-level systems needs $J\,T \geq \pi/4$, with J the norm of the interaction on the full d-level space. The same holds for every gate only locally equivalent to CNOT, such as IBM's echoed cross-resonance (ECR): such a gate maps some product vector to*

> *$(P \otimes Q)$ applied to a Bell vector, local unitaries do not change the overlap with product vectors, and so $JT \geq \pi/4$.*

Lean · ours, in part **show**

the Bell-vector bound and its extension to (P⊗Q)·Bell, which covers every gate only locally equivalent to CNOT (ECR); the CNOT consequence itself is the Bell bound applied to |+⟩|0⟩

- `QSL.Qudit.pi_div_four_le_entangling_time_fin`
- `QSL.Qudit.pi_div_four_le_entangling_time_local`

✅ Checked when this page was built: QuditEntangling.lean, QuditEntanglingShift.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — pi_div_four_le_entangling_time_fin, pi_div_four_le_entangling_time_local: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean QuditEntangling.lean QuditEntanglingShift.lean` (SHA-256 of the compiled file: `f4a5538fe9427bac…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** a consequence of the Bell bound by the Schmidt argument (CNOT maps |+⟩|0⟩ to the Bell state); with pre-entangled ancillas the argument does not apply.

↳ Not used by another result of the paper.

🛡️ **Evidence & scope**

- **Program:** independent numerical check of the overlap constant: m_d = 0.707107 (d = 2) and 0.707105 (d = 3) for the largest overlap of a CNOT output with product vectors, over 45 starts, consistent with $1/\sqrt{2}$; a maximization from finitely many starts is a check, not a certificate (quoted from the paper: the re-run for this blueprint reproduced m_2 = 0.707107 and the embedded-start control, but the 45-start m_3 search did not finish in 10 minutes). `cnot_m3.py`

Theorem §II-B. ✅ Lean **(Robust speed limits)✓**

> *If the evolution maps a product vector to a state whose overlap with the Bell vector satisfies $|\langle \mathrm{Bell}, \Phi\rangle|^2 \geq 1-\varepsilon$, then $J\sum_k \tau_k \geq \pi/4 - \arcsin\sqrt{\varepsilon}$. For a two-level transfer with infidelity at most ε, $\Omega_{\max} T \geq 2\arccos\sqrt{\varepsilon}$.*

Lean · ours **show**

- `QSL.Qudit.pi_div_four_sub_le_entangling_time_approx`
- `QSL.Qudit.pi_div_four_sub_le_entangling_time_shift_approx`
- `QSL.Gate.two_arccos_sqrt_le_time_approx`

✅ Checked when this page was built: ApproxSpeedLimits.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — pi_div_four_sub_le_entangling_time_approx, pi_div_four_sub_le_entangling_time_shift_approx, two_arccos_sqrt_le_time_approx: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean ApproxSpeedLimits.lean` (SHA-256 of the compiled file: `a1ac6d198aa29365…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** a triangle inequality in the Fubini–Study angle; «new as stated» (Table I). ε is the infidelity of one input/output pair, not an average gate infidelity. The two-

level transfer bound is tight ($H = (\Omega/2)\sigma_x$).

↳ Used in: §VII-B (p. 36), §VIII (p. 40)

Proof

Formalized in Lean (see the linked declarations).

3 Data and Conventions

Observation §III-C. (The drive convention)

The factor c is not documented in the snapshots. We fixed it before computing any gap, with the pre-specified rival H3 ("a different convention, factor 2 or 2π"). The independent datum is the calibrated X pulse itself: its rotation angle must be π. Across the 32 backends with pulse definitions, `omegad` $\cdot\,|\mathrm{amp}| \cdot$ area $= 1.148\,\pi$ with $c = 1$ and $2.296\,\pi$ with $c = 2$ (median, with a simple discretization of the area). We adopt $c = 1$: the conventions with a factor 2 or 2π are excluded. The residual ≈ 1.15 has an exact explanation in the pulse normalization (next node).

🆕 **Classical and new:** pre-registered rival H3 («a different convention, factor 2 or 2π») excluded; the residual ≈ 1.15 is explained by the pulse normalization (next node).

↳ Used in: §III-C (p. 18), §III-C (p. 18), §III-D (p. 19), §IV-A (p. 19), §IV-C (p. 21)

💡 **In plain words (commentary, not part of the paper)**

What it says. IBM publishes a drive-strength parameter, `omegad`, without saying how it converts to a Rabi frequency (a factor c of 1, 2 or 2π is possible). Every gap in the paper depends on that factor, so the paper fixes it before computing any gap, using one independent datum: IBM's calibrated X pulse must rotate the qubit by exactly π.

What was found. Over the 32 backends with pulse definitions the rotation comes out as $1.148\,\pi$ with $c = 1$ and $2.296\,\pi$ with $c = 2$. Only $c = 1$ is close to π, so $c = 1$ is adopted and the factor-2 and factor-2π conventions of the rival H3 are excluded. The leftover $\approx 15\%$ is now explained (next node): it is the ratio of two pulse areas, which tends to $1/(1 - e^{-2}) = 1.15652$. The snapshots do not say which of the two waveforms reaches the qubit, so the single-qubit figures at the drive actually used are given with both readings (shape factor 1.87 or 2.16).

What this is not. It is a measurement on data, so this node is grey: no Lean, no proof. The evidence is the programs named below, run by the author; the numbers are the paper's, from `pulsos_x.py`, `identidad_omegad.py`, `identidad_calibracion.py` and, for the exact identity, `cruces_area_cz.py`.

🛡️ **Evidence & scope**

- **Program:** the calibrated X pulse of the 32 backends with pulse definitions: median θ/π = 1.148 (c = 1), 2.296 (c = 2); n = 32 (output saved in lab/salidas/pulsos_x.txt). `pulsos_x.py`
- **Program:** with a simple discretization the residual depends on the Gaussian width (1.1428 for σ = 24, 1.1523 for σ = 80). `identidad_omegad.py`

- **Program:** with Qiskit's own convention (samples at the midpoints, zero one sample beyond each end) the dependence almost disappears: 1.1552–1.1565 for σ = 24–640. `identidad_calibracion.py`, `identidad_calibracion_salida.txt`
- **Program:** three-level check of the drive convention. `convencion.py`

Observation §III-C. (The pulse-area identity)

The residual ≈ 1.15 has an exact explanation in the pulse normalization. With $g(x) = \exp[-(x - N/2)^2/(2\sigma^2)]$, $N = 4\sigma$ and $S = \sum_{n=0}^{N-1} g(n + \frac{1}{2})$, let $A_{\text{rec}} = dt\,[S - N\,g(-\frac{1}{2})]$ be the area of the sampled Gaussian with the edge value subtracted but *without* renormalizing the peak, and $A_Q = dt\,[S - N\,g(-1)]/[1 - g(-1)]$ the area of Qiskit's lifted Gaussian. Then `omegad` $\cdot\,|\text{amp}| \cdot A_{\text{rec}} = \pi$ on all 777 qubits with pulse definitions, to $1.2 \cdot 10^{-7}$; the residual is A_Q/A_{rec}, which tends to $1/(1 - e^{-2}) = 1.15652$ and reproduces, without fitting, both the width dependence of a simple discretization (1.1428, 1.1482, 1.1523 for $\sigma = 24, 40, 80$) and the values 1.1552–1.1565 of Qiskit's convention. The check covers all qubits. So `omegad` is consistent with a drive whose peak is $(1 - e^{-2})|\text{amp}|$ in units of `omegad`; under Qiskit's normalization the model would over-rotate by 1.1565. The snapshots do not say which waveform reaches the qubit. The full-amplitude ratio below uses `omegad` only and does not depend on this; the peak-drive figures do, and we give both readings.

> 🆕 **Classical and new:** checked over all 777 qubits. The snapshots do not say which waveform reaches the qubit: the full-amplitude ratio uses omegad only and does not depend on it, the peak-drive figures do, and the paper gives both readings (shape factor 1.87 or 2.16; Ω_peak/|δ| 0.078 or 0.0907).

↳ Used in: §IV-C

Evidence & scope

- **Program:** omegad·|amp|·A_rec/π − 1 over 777 qubits: max |·| = 1.18·10⁻⁷, median −1.07·10⁻⁹; 1/(1−e⁻²) = 1.1565176; the area ratio A_Q/A_rec = 1.1551951875 (σ = 24), 1.1558005158 (σ = 40), 1.1561888528 (σ = 80), 1.1564799045 (σ = 640) (regenerated output of the package). `cruces_area_cz.py`, `cruces_area_cz_salida.txt`
- **Program:** the width dependence of a simple discretization (1.1428, 1.1482, 1.1523 for σ = 24, 40, 80) and the values 1.1552–1.1565 of Qiskit's convention are reproduced without fitting. `identidad_calibracion.py`, `identidad_calibracion_salida.txt`

Observation §III-C. (omegad is derived)

For equal σ, the product `omegad` · amp is identical to 9 digits across 27 backends. Hence `omegad` is *not* an independent measurement; IBM derives it from the π-pulse calibration. Any correlation between gate duration and `omegad` is therefore partly circular.

↳ Used in: §IV-B (p. 21)

Evidence & scope

- **Program:** for equal σ, omegad·amp is identical to 9 digits across 27 backends. `identidad_omegad.py`

Observation §III-D. (Exclusions and falsifier)

The pre-registered falsifier was a ratio $\tau_X/\tau_{\mathrm{QSL}}<1$ on any qubit, impossible for an exact X under a correct convention (an approximate gate with trace error e allows down to $(2/\pi)\arcsin(1-e)$, i.e. 0.991 for $e=10^{-4}$). It fired: 254 qubits came out faster than their own limit. They belong to the snapshots of `brussels`, `prague` and `strasbourg`, whose sample time $dt \geq 1$ ns differs from that of every other snapshot (0.22 ns) and suggests placeholder values (we have no documentation of their provenance); these backends are excluded, as are qubits reporting zero anharmonicity; the exclusion was decided after the falsifier fired, and we report it as such. Controls: a hand computation on `armonk` reproduces the script (ratio 2.631), and changing the unit of `omegad` moves the median by exactly 2π. One time-series snapshot (`toronto`, 2021-03-15, median two-qubit duration 6315 ns, ratio ~ 40) is an outlier, probably a degraded calibration; it is kept in the time-series table, and removing it changes the 2021 two-qubit median from 2.78 to 2.74 (`toronto_sensibilidad.py`).

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** the single-qubit sweep with its falsifier, exclusions and controls (armonk by hand 2.631; unit change moves the median by exactly 2π); the blueprint recomputes the sweep statistics from barrido.csv. `barrido.py`, `lab/comprueba_numeros.py`
- **Program:** the toronto 2021-03-15 outlier (median two-qubit duration 6315 ns, ratio 40) is kept in the time series; removing it moves the 2021 two-qubit median from 2.78 to 2.74 and the single-qubit one not at all. `toronto_sensibilidad.py`, `toronto_sensibilidad_salida.txt`
- **Program:** the falsifier with tolerance: an approximate X gate with trace error e = 10^{-4} allows ratios down to (2/π) arcsin(1−e) = 0.990997. `tercera_revision.py`, `tercera_revision_salida.txt`

4 Single-Qubit Gates

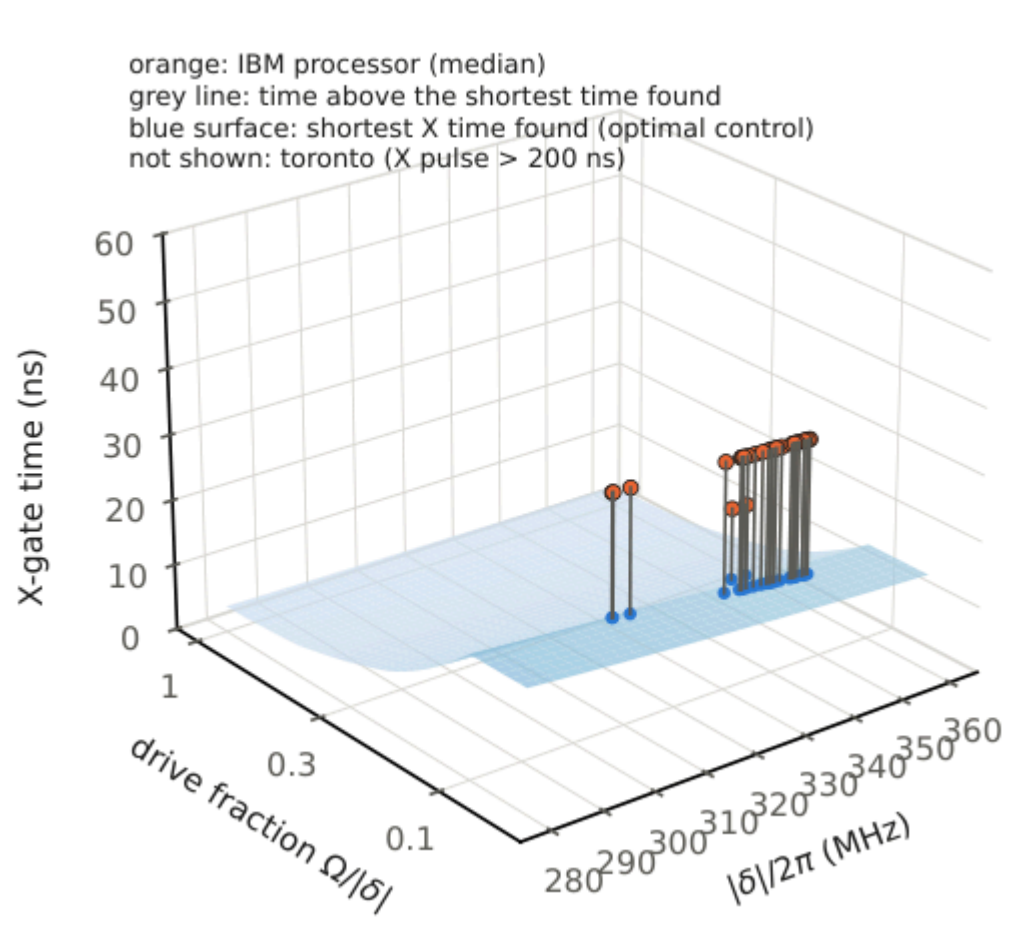
Observation §IV-A. (The single-qubit gap)

Over 41 backends and 1646 qubits, the X gate lasts a median $\tau_X/\tau_{\mathrm{QSL}}=11.30$ times its two-level speed limit at full drive amplitude ($|D|=1$, $\tau_{\mathrm{QSL}}=\pi/$`omegad`), 5th–95th percentile $[7.19, 15.54]$, minimum 2.37. The $\sqrt{X}$ gate has the same duration and half the rotation, so its median ratio is 22.61. Qubits of one backend share a duration, so they are not independent: the median of the 41 per-backend medians is 11.37, and a bootstrap that resamples whole backends gives a 95% interval $[11.02, 11.65]$ for the median. Against the leakage time scale $2\pi/|\delta|$, the X gate lasts a median 15.45 periods $[10.90, 18.70]$. IBM does not drive at that amplitude: the next nodes separate the part of the gap set by the drive actually used from the part set by amplitude headroom.

🆕 **Classical and new:** the diagnostic: to the paper's knowledge no prior work applies a proof-assistant-certified speed limit to public calibration data with a per-gate gap (a search that finds nothing does not prove novelty).

↳ Used in: §IV-C, §IV-B (p. 21), §VI (p. 26), §VII-A (p. 28), §XI (p. 45)

🖼️ **Figure 1**



Where IBM single-qubit gates sit relative to the model. Blue surface: shortest X-gate time found by optimal control on a 4-level transmon (Table ??) as a function of the anharmonicity and of the drive fraction $\Omega/|\delta|$. Orange: one point per IBM processor with calibrated pulses (median anharmonicity, median drive fraction $\Omega_{\text{peak}}/|\delta|$, X duration); the grey line is the time above the shortest time found at the same drive fraction. `toronto` (2560-sample X pulse) is not shown but is included in the medians. The surface for $r \geq 0.8$ uses non-converged 4-level values. Section ??. Interactive version.

💡 **In plain words (commentary, not part of the paper)**

What it says. Take IBM's own description of a transmon and ask: how fast could a single X gate be if the drive were used at its full strength, with a square pulse? That is the speed limit $\tau_{\text{QSL}} = \pi/\Omega_{\max}$. Compare it with the duration IBM actually calibrates. Over 1646 qubits on 41 backends the real X gate lasts a median 11.30 times longer (5th–95th percentile 7.19 to 15.54; the best qubit is at 2.37). The $\sqrt{X}$ gate lasts the same time but needs half the rotation, so its median ratio is 22.61. Measured against the leakage time scale $2\pi/|\delta|$ the X gate lasts 15.45 periods.

Where the numbers come from. Recomputed from the archived output `barrido.csv` by `lab/comprueba_numeros.py` of this blueprint (11.304, 7.186, 15.540, 2.375, 22.608,

15.449; each with a control that must fail). The figure is the paper's Figure 1: the lab below makes it live.

What it means, and what it does not. The paper reads the gap as set by engineering choices (pulse shape, a duration fixed in samples, a low amplitude), not by the speed limit or by the anharmonicity. IBM does not drive at full amplitude: at the drive it uses, the gap is about 2 (shape factor 1.87, or 2.16 depending on which waveform reaches the qubit). The model is IBM's, not a measurement.

Evidence & scope

- **Program:** 41 backends, 1646 qubits: median τ_X/τ_QSL 11.30, 5th-95th percentile [7.19, 15.54], minimum 2.37; √X median 22.61; 15.45 leakage periods; recomputed from barrido.csv with controls that must fail (11.304, 7.186, 15.540, 2.375, 22.608, 15.449). `barrido.py`, `lab/comprueba_numeros.py`
- **Program:** qubits of one backend share a duration, so they are not independent: the median of the 41 per-backend medians is 11.37 and a bootstrap that resamples whole backends gives a 95

Evidence & scope

- **By hand:** τ_X/τ_QSL = duration·dt·omegad/π; since omegad·|amp|·A_rec = π, the ratio is exactly duration·dt/(|amp|·A_rec) ≈ 2.16/|amp|; 2.16 = duration over A_rec = the pulse-shape factor 1.87 (a 4σ Gaussian with its edge removed against a square pulse of the same peak) times 1/(1−e⁻²); 1/amp ≈ 5 (IBM calibrates the π pulse at amp ≈ 0.2). Quoted from the paper; no separate script output reproduces the decomposition

Observation §IV-B. (Pre-registered hypotheses)

Before computing we wrote three rivals with predicted ranges: H1, amplitude-limited gates at 1–2× the limit; H2, leakage-limited gates, $\tau_X \propto 1/|\delta|$, ratio 2–6, uncorrelated with `omegad`; H3, a convention artefact. Outcome: H1 is rejected against the full-amplitude limit (the gates are at ≈ 11×); against the peak drive actually used the ratio is ≈ 2.16, just above H1's range, which did not specify the drive; with the peak $(1 - e^{-2})|\text{amp}|$ consistent with `omegad` it is 1.87, inside that range, so this comparison depends on the reading, while the full-amplitude one does not; H2 has no support (R^2 of $\log \tau_X$ on $\log(1/|\delta|)$ is 0.087, and the gates last ≈ 15 leakage periods); the factor-2 and factor-2π conventions of H3 are excluded by the π-pulse check, and the remaining 15% is the lifted-Gaussian normalization. The R^2 on $\log(1/\texttt{omegad})$ is 0.692 (shuffled control 0.001), but because `omegad` is derived from the same pulse's amplitude and width this is partly circular and is not independent physical evidence.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** R² of log τ_X on log(1/|δ|) = 0.087, on log(1/omegad) = 0.692, shuffled control 0.001 (printed by barrido.py; quoted from RESULTADOS.md, not recomputed here); the hypotheses and their predicted ranges were written before computing. `barrido.py`, `ficha_ideas.toml`

Observation §IV-C. (A cap near 0.1)

The calibrated peak Rabi frequency sits at a nearly fixed fraction of the anharmonicity: over 750 qubits, $\Omega_{\text{peak}}/|\delta|$ has median 0.0907 and 10th–90th percentile $[0.0607, 0.0992]$, i.e. a cap near 0.1, taking the peak as `omegad` $|\text{amp}|$; with the peak $(1 - e^{-2})$ `omegad` $|\text{amp}|$ consistent with `omegad` the median is 0.078. We read this as IBM's design rule, not a physical limit.

↳ Used in: §IV-C (p. 22)

Evidence & scope

- **Program:** 750 qubits: Ω_peak/|δ| median 0.0907, 10th-90th percentile [0.0607, 0.0992] (output saved in lab/salidas/cruce_formulas.txt). `cruce_formulas.py`

Observation §IV-C. (Per-qubit durations)

Since all qubits of a backend share one duration, one may ask how much would be gained by giving each qubit the shortest duration compatible with that cap ($c = 0.099$, the observed 90th percentile), keeping the pulse shape (750 qubits on 31 backends). The median per-qubit gain is only $\times 1.09$ ($\times 1.07$ if a layer waits for its slowest qubit): the cap binds almost every qubit alike. The backend `cairo` is a natural experiment: its common X duration is 21.3 ns where the cap would require a median of 31.6 ns, i.e. it runs at a drive fraction near 0.15, and works.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** 750 qubits on 31 backends: τ_i/τ_common median 0.92 [p10 0.61, p90 1.00], gain ×1.09 (×1.07 if a layer waits for its slowest qubit); cairo: common 21.3 ns against a median 31.6 ns required by the cap (archived output). `mejora_por_cubit.py`, `mejora_por_cubit_salida.txt`

5 Two-Qubit Gates

Observation §V-A. (The pairs)

From the snapshots we extract every coupled pair with a coupling j_q in the Hamiltonian variables and a calibrated two-qubit gate: 1758 pairs in 43 backends (1019 ECR, 739 CX). The median coupling is $j_q = 0.01268$ rad/ns ($g/2\pi \approx 2.0$ MHz), the median detuning 115 MHz ($\approx 60\, j_q$), and the median gate duration 548 ns (5th–95th percentile $[277, 698]$ ns).

↳ Used in: §V-B (p. 22), §V-D (p. 24)

Evidence & scope

- **Program:** 1758 pairs, 43 backends, 1019 ECR and 739 CX, median j_q 0.01268, median duration 548 ns; recomputed from pares.csv. `pares.py`, `lab/comprueba_numeros.py`

Observation §V-B. (The two-qubit gap)

In the qubit subspace the IBM coupling is $j_q(\sigma_i^+\sigma_j^- + \text{h. c.}) = (j_q/2)(XX + YY)$, of operator norm j_q. *Literature bound for the gate.* With unbounded local control, no

leakage, and an XY coupling, the minimal time of a CNOT-class gate (CNOT, CZ, ECR) is $\pi/(2j_q)$. This follows from Howard *et al.*, Appendix C, and from the majorization theory of Vidal–Hammerer–Cirac. The detuning is a local term absorbed by the unbounded local control, and bounded drives are a subset of unbounded ones, so within the qubit subspace the real duration cannot beat it [our inference, not stated in that form in the sources]. *Our certified bound for state preparation.* The Lean theorem bounds the time to *create* a Bell state from a product state: $\pi/(4J)$, attainable. A full CNOT on an XY coupling needs twice that, which is why the literature bound is the stronger one for gates. Ours is the one that is machine-checked, and it survives extra transmon levels: with $|2\rangle$ included the norm of the exchange term grows ($\|a^\dagger b + ab^\dagger\| = 1, 2, 3.4641$ for $d = 2, 3, 4$), and the CNOT bound becomes $\pi/(4J)$ with that larger J. With leakage levels allowed, the only rigorous constant in the literature is for iSWAP; for CNOT/ECR with $|2\rangle$ no constant was found in the sources read. Ours, $\pi/(8j_q)$ for $d = 3$, is not known to be tight. Against the literature bound the median ECR/CX lasts 4.22 times its limit.

> 🆕 **Classical and new:** the bound is from the literature: Howard et al., Appendix C, and Vidal–Hammerer–Cirac (not in Lean); the step «the real duration cannot beat it» is our inference, not stated in that form in the sources (the paper says so).

↳ Used in: §V-C (p. 24), §VI (p. 26), §XI (p. 45)

💡 In plain words (commentary, not part of the paper)

What it says. For a two-qubit gate of the CNOT family (CNOT, CZ, ECR), with an exchange coupling j_q between the two transmons and free local control, the shortest possible time is $\pi/(2j_q)$. This is a bound from the literature (Howard *et al.*, Appendix C, and Vidal–Hammerer–Cirac), not a Lean theorem of ours. At the median coupling $j_q = 0.01268$ rad/ns that is 124 ns; the median calibrated gate lasts 548 ns.

The number. Over 1758 coupled pairs on 43 backends (1019 ECR, 739 CX) the median gate lasts 4.22 times the bound. Recomputed from `pares.csv` by `lab/comprueba_numeros.py`: 4.215, with the pair count, the ECR/CX split, the median j_q and the median duration all matching the paper.

Which bound. The paper keeps two apart: the stronger literature bound for whole gates (4.22), and the one proved in Lean, which bounds the time to *create* a Bell state, $\pi/(4J)$, half as large; against it the ratio is exactly twice, ≈ 8.7. They answer different questions and both are correct. Both assume the qubit subspace, $\|H_{\text{int}}\| = j_q$. With the second transmon level the norm grows to $2j_q$, the certified time bound halves to $\pi/(8j_q)$ and the certified ratio doubles to about 16.9: the bound becomes weaker, which says nothing about how much of that room is reachable.

🛡️ Evidence & scope

- **Literature:** with unbounded local control, no leakage and an XY coupling the minimal time of a CNOT-class gate is π/(2j_q) (Howard et al., Appendix C, with g_H = j_q/2; Vidal–Hammerer–Cirac majorization)
- **Program:** median ratio 4.22 over 1758 pairs (recomputed from pares.csv: 4.215). `pares.py`, `lab/comprueba_numeros.py`

Observation §V-C. (The certified ratio)

Against our certified chord constant $\sqrt{2-\sqrt{2}}/j_q$ the median ratio is 8.7; the sharp Lean constant $\pi/4$ lowers it by 2.6% to 8.4, and since $\pi/(4j_q)$ is exactly half of $\pi/(2j_q)$ the certified ratio is exactly twice the literature ratio. Both are correct; they bound different tasks. Both assume $\|H_{\text{int}}\| = j_q$, i.e. the qubit subspace. With the second transmon level the norm grows to $2j_q$, the certified time bound halves to $\pi/(8j_q)$ and the certified ratio doubles to ≈ 16.9: the bound becomes weaker, which says nothing about how much of that room is reachable. Our $\pi/(8j_q)$ for $d = 3$ is not tight; Section VII sharpens it to $0.4511/j_q$ for the CZ.

Lean · ours, in part **show**

the two bounds; the ratios (8.7 chord, 8.4 sharp, and ≈ 16.9 with the second level) are computations on vendor data, not Lean statements

- `QSL.TwoQubit.entangling_time_lower_bound`
- `QSL.Qudit.pi_div_four_le_entangling_time_fin`

✅ Checked when this page was built: QuditEntangling.lean, TwoQubitEntangling.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — entangling_time_lower_bound, pi_div_four_le_entangling_time_fin: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean QuditEntangling.lean TwoQubitEntangling.lean` (SHA-256 of the compiled file: `d1b9ef08f2a9b3aa…`; a source page on this site may show it with comments edited, under its own hash).

↳ Used in: §VII-B (p. 33)

Evidence & scope

- **Program:** median of duration·j_q/√(2−√2) over the 1758 pairs: 8.65 (paper: 8.7). `pares.py`, `lab/comprueba_numeros.py`

Observation §V-C. (Numerical controls)

Numerical controls of the certified bound: the Duhamel inequality has 0 violations in 20 000 random pairs, with maximum ratio 0.99998 (the constant 1 is sharp), and the bound has 0 violations in 3000 evolutions with random local control.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** the Duhamel inequality: 0 violations in 20 000 random pairs, maximum ratio 0.99998; the bound: 0 violations in 3000 evolutions with random local control; achievable time π/(4J) = 0.7854/J, bound 2.6

Observation §V-D. (Cross resonance and the first-order formula)

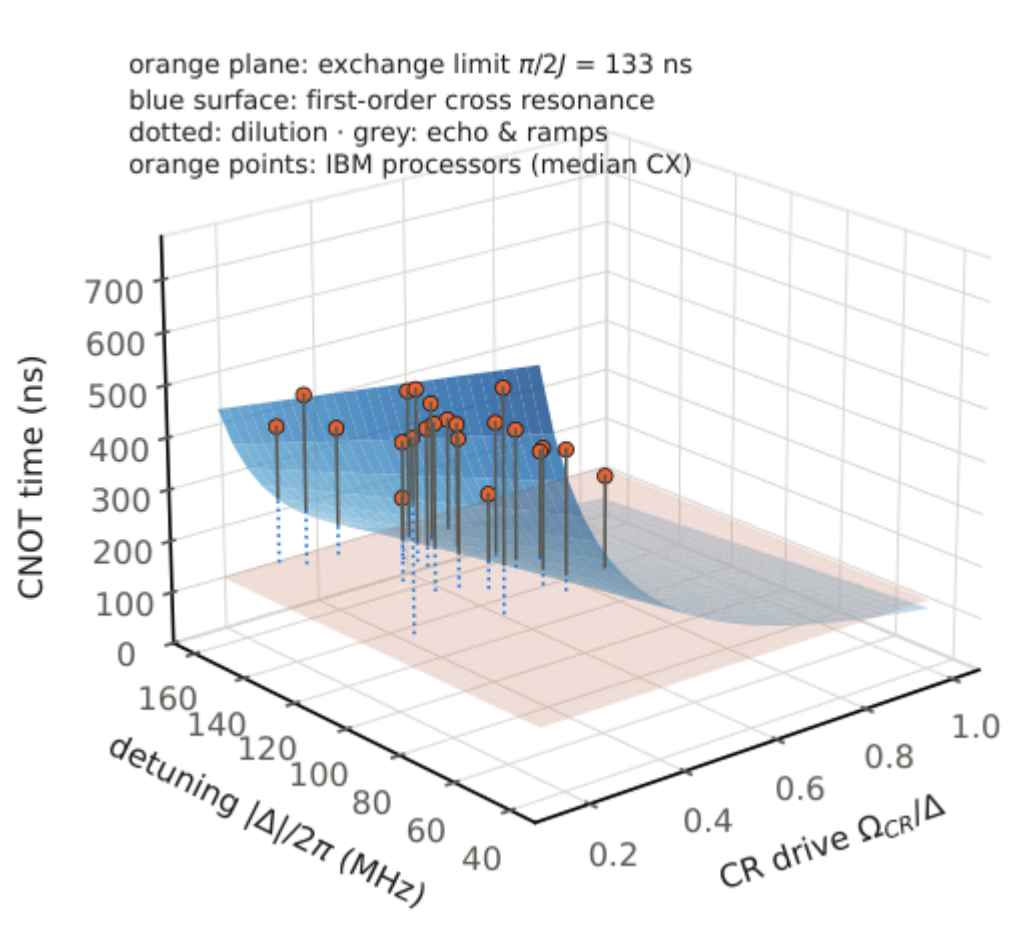
IBM's CX is built from echoed cross-resonance (CR) pulses. To first order the CR drive of amplitude Ω on the control, at detuning Δ, produces a ZX rate $\nu_{ZX} = -(J\Omega/\Delta)\,\delta_c/(\delta_c + \Delta)$, and a CNOT needs $\theta_{ZX} = \nu_{ZX} t_{\text{CR}} = \pi/2$. On 1212 calibrated CX gates in 29 backends the median of $\theta/(\pi/2)$ is 1.109, 10th–90th percentile $[0.870, 1.472]$, with 59% in $[0.8, 1.2]$. The median lies 11% above the target, with a wide spread. The calibrated drive sits at a nearly fixed fraction of the detuning: $|\Omega_{\text{CR}}/\Delta|$ has median 0.451. The two gaps share one pattern: the control runs at a fixed

fraction of an energy splitting it must not excite—$\Omega_{\text{peak}} \approx 0.09|\delta|$ for one qubit, $\Omega_{\text{CR}} \approx 0.45\Delta$ for two. The calibrated drives lie outside the stated range of the perturbative expansion (53% of the gates have $\Omega/2\pi > 50$ MHz); with the third-order term the median falls to 0.898 ($[0.659, 1.105]$), and a numerical block diagonalization gives 0.967 ($[0.795, 1.159]$, 81% in $[0.8, 1.2]$). Flipping the sign of Δ moves the median away from 1 (0.664), as a control should. One convention is unresolved: the Qiskit control channel of the pair (c, t) is weighted by the drive strength of the *target*, whereas we use that of the control; with the former the first-order median is 1.089.

> 🆕 **Classical and new:** the first-order expression is Magesan–Gambetta (2020), Eq. (4.26); first order means no ZZ, IX or classical crosstalk, and the angles are not checked against a time-dependent simulation of the calibrated sequence. Higher-order checks below are effective Hamiltonians, not that simulation.

↳ Used in: §VII-B (p. 32)

🖼️ **Figure 2**



Two gaps in the two-qubit gate. Orange plane: exchange speed limit $\pi/(2J) = 133$ ns at the median coupling of the 29 processors with pulse data ($J = 0.0118$ rad/ns; the median over all 1758 pairs of Section ?? is 0.01268). Blue surface: CNOT time allowed by first-order cross resonance, $t = (\pi/2)/\nu_{ZX}$, as a function of the drive fraction Ω_{CR}/Δ and the detuning. Orange points: one per IBM processor with calibrated CX pulses (median drive fraction, detuning and CX duration). Dotted: dilution by the cross-resonance mechanism (plane to surface); grey: the remainder (surface to point), from echo pulses and ramps and from the error of the first-order model. Median J and anharmonicity. Interactive version.

Evidence & scope

- **Program:** 1212 calibrated CX gates in 29 backends: median θ/(π/2) = 1.109 [p10 0.870, p90 1.472], 59
- **Program:** the calibrated drives lie outside the stated range of the perturbative expansion (53
- **Numerical illustration (not a proof):** the lab below redraws Figure 2: the first-order surface, the exchange-limit plane and the 29 processors. `lab/qsl_diagnostics_lab.py`

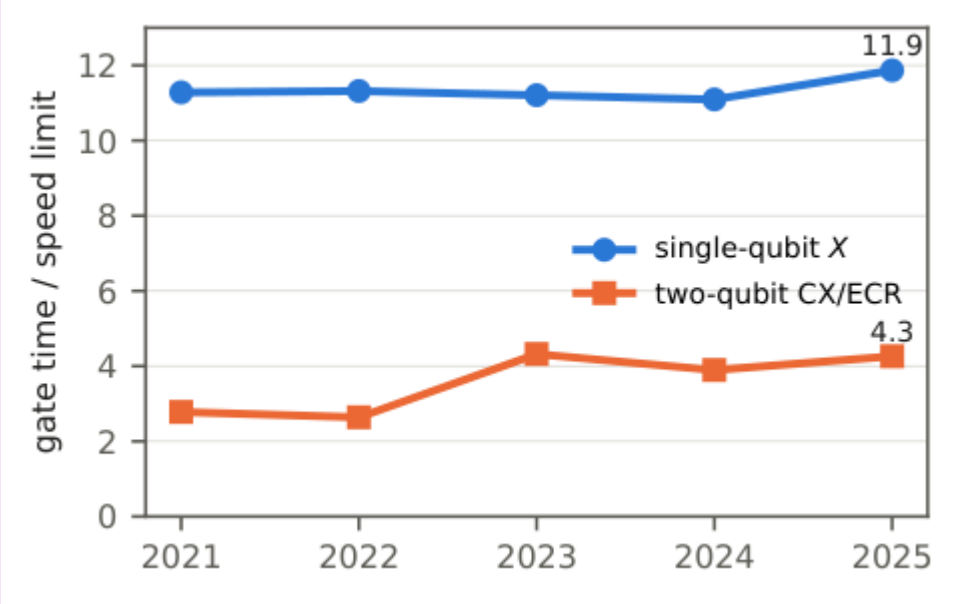
6 Time Series 2021–2025

Observation §VI. (Median gaps by year)

Median gaps by year (56 usable snapshots, 41 backends). Two-qubit ratio against $\pi/(2j_q)$. Single qubit $\tau_X/\tau_{\mathrm{QSL}}$: 11.28 (2021), 11.32 (2022), 11.21 (2023), 11.10 (2024), 11.87 (2025). Two qubits $\tau_{2q}/(\pi/2j_q)$: 2.78, 2.64, 4.32, 3.89, 4.26. The ratio stays at ≈ 11 over five years for a single qubit: a design constant. For two qubits it grew from 2.78 (2021) to 4.26 (2025). Snapshots per year: 27, 4, 2, 16, 7; the 2022 and 2023 medians rest on 4 and 2 snapshots respectively. One time-series snapshot (`toronto`, 2021-03-15, median two-qubit duration 6315 ns, ratio ~ 40) is kept in the table; removing it changes the 2021 two-qubit median from 2.78 to 2.74 and the single-qubit one not at all.

↳ Used in: §VI

Figure 3



Per-year median of the snapshot medians: single-qubit ratio frozen near 11, two-qubit ratio growing with the gate duration. Interactive version, split by gate or processor size and followed processor by processor.

In plain words (commentary, not part of the paper)

What it says. Repeating the two gaps on Qiskit releases spanning 2021 to 2025 (13 releases, 101 unique (backend, date) snapshots, 56 usable from 41 backends), the single-qubit ratio does not move and the two-qubit ratio grows.

	2021	2022	2023	2024	2025
$\tau_X/\tau_{\mathrm{QSL}}$	11.28	11.32	11.21	11.10	11.87
$\tau_{2q}/(\pi/2j_q)$	2.78	2.64	4.32	3.89	4.26

These are the paper's Table IV; the blueprint recomputes every entry from `serie.csv` (median of the snapshot medians, per year) with `lab/comprueba_numeros.py`, and each agrees to two decimals.

Reading. The single-qubit ratio is a design constant. The two-qubit ratio rose because the *duration* rose, not because couplings fell: $g/2\pi \approx 1.9$–2.1 MHz in every processor family, and within a processor j_q is identical at every date (a model value carried over, not re-measured). Caveat from the paper: the 2022 and 2023 medians rest on only 4 and 2 snapshots.

Evidence & scope

- **Program:** median of the snapshot medians per year, 56 usable snapshots: all ten table entries recomputed from serie.csv to two decimals. `serie_temporal.py`, `lab/comprueba_numeros.py`

Evidence & scope

- **Program:** j_q identical at every date within a processor; Eagle ECR median 562 ns against Falcon CX ≈ 377 ns; recalibrations lengthen (bogota 366→544, perth 380→491, lagos 299→452). `serie_temporal.py`, `serie_porque.py`

Observation §VI. (Lengthening and error)

Across the 80 snapshots with two-qubit durations and errors, Spearman's correlation between median two-qubit duration and median gate error is $+0.13$ ($p = 0.23$). For the same pair lengthened between two dates (71 cases), the error *rises* (median change in log error $+0.21$; it falls in only 31%); for pairs shortened (40 cases) it falls. These comparisons are observational: other calibration changes and drift happen at the same dates.

↳ Used in: §VI (p. 27)

Evidence & scope

- **Program:** Spearman(duration, error) = +0.135 (p = 0.231) over 80 snapshots; the same pair lengthened between two dates: 71 cases, median change in log error +0.214, falls in 31

Observation §VI. (A methodological lesson)

We then asked whether lengthened pairs had worse qubits. Pooled across dates, T_1 of lengthened pairs fell more than that of unchanged pairs (Mann–Whitney $p = 0.0017$). Under a stratified control (same backend, same dates; 18 strata, 84 lengthened pairs) the

effect vanished (Wilcoxon $p = 0.90$ for T_1, $p = 0.99$ for T_2); a pooled frequency-shift effect ($p = 4 \cdot 10^{-12}$) vanished likewise. The pooled effects are consistent with confounding by epoch: lengthenings concentrate in backends and periods in which the whole chip's T_1 fell. We retracted the reading that lengthened pairs had degraded qubits: after controlling for backend and date the data do not support pair-specific degradation. This does not show what does drive lengthening. Any cross-date effect in calibration snapshots should be tested stratified by backend and date before it is read.

> 🆕 **Classical and new:** a retraction by the authors of their own «degraded pairs» reading; the lesson: test any cross-date effect in calibration snapshots stratified by backend and date.

↳ Not used by another result of the paper.

🛡️ **Evidence & scope**

- **Program:** pooled Mann-Whitney p = 0.0017 (T1) against stratified Wilcoxon p = 0.90 (T1) and 0.99 (T2) over 18 strata and 84 lengthened pairs (output saved in lab/salidas/degradacion.txt). `degradacion.py`
- **Program:** pooled frequency-shift effect p = 4·10⁻¹² against stratified ×1.0 (greater in 49

7 How Far the Model Allows

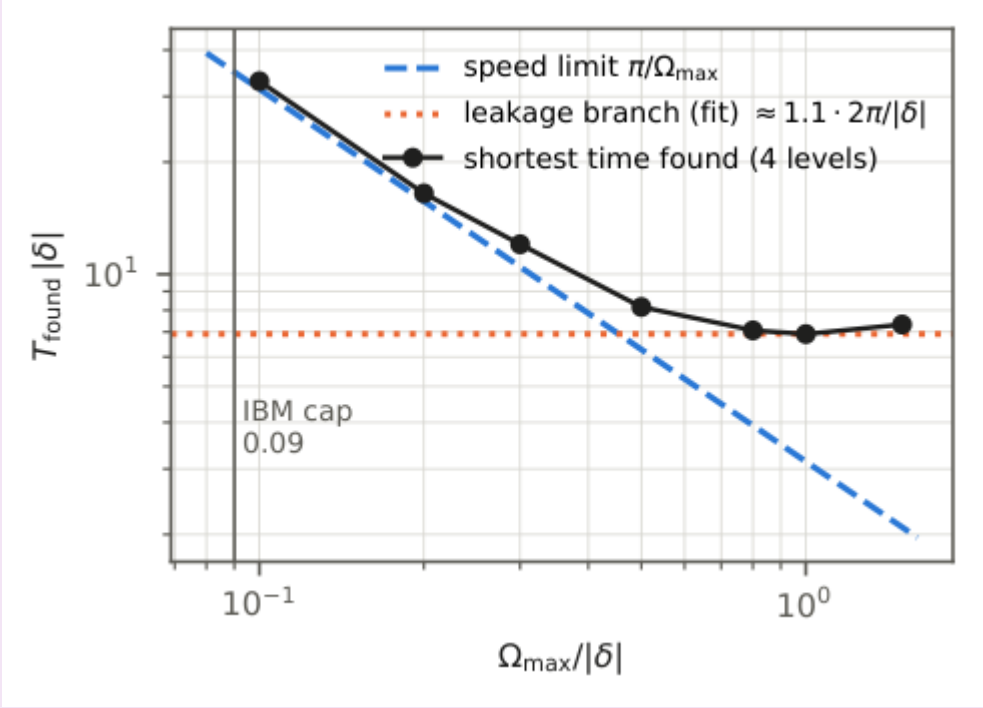
Observation §VII-A. (The single-qubit law)

We ran optimal control on a 4-level Duffing transmon in units $|\delta| = 1$, with a piecewise-constant complex drive bounded by $|\Omega| \le r|\delta|$, and searched, on a grid of times, for the shortest time at which an X gate with error $<10^{-4}$ including leakage is found. The two-level control returns $T = \pi/\Omega_{\max}$ exactly. The results follow $T_{\text{found}} \approx \max\big(\pi/\Omega_{\max},\ \sim 1.1 \cdot 2\pi/|\delta|\big)$, a speed-limit branch for $r \lesssim 0.3$ and a leakage-dominated branch for $r \gtrsim 0.8$. This is an empirical fit to a coarse grid, not a theorem; it misses the values found at $r = 0.3$–0.5 by 13–16%, where neither branch dominates. The entries are feasible times found, i.e. upper bounds on the true minimum: a larger cap admits every pulse of a smaller one, so the true minimum cannot increase with r, and the value at $r = 1.5$ above that at $r = 1.0$ shows that the search is local. Shortest time found (units $|\delta| = 1$; ratio to $\pi/\Omega_{\max}$): $r = 0.1$: 33.0 (1.05); 0.2: 16.5 (1.05); 0.3: 12.0 (1.15); 0.5: 8.2 (1.30); 0.8: 7.1; 1.0: 6.9; 1.5: 7.3.

> 🆕 **Classical and new:** an empirical fit to a coarse grid, not a theorem (the paper). The entries are feasible times found, i.e. upper bounds on the true minimum; the value at r = 1.5 above that at r = 1.0 shows that the search is local. Joining the two branches in a certified bound is an open question.

↳ Used in: §VII-A (p. 30)

Figure 4



Shortest X-gate time found by optimal control on a 4-level transmon (error $<10^{-4}$, leakage included) against the drive bound: speed-limit branch $\pi/\Omega_{\max}$ and leakage-dominated branch $\approx 1.1 \cdot 2\pi/|\delta|$ (empirical fit). The vertical line marks the cap observed in IBM calibrations. Interactive version.

In plain words (commentary, not part of the paper)

What it says. Suppose you may drive a transmon as hard as a fraction $r = \Omega_{\max}/|\delta|$ of its anharmonicity (driving harder excites the third level, "leakage"). How short can an X gate be, with the best possible pulse shape? The paper ran numerical optimal control on a 4-level model, asking for error below 10^{-4} including leakage. In units $|\delta| = 1$ the shortest times found (feasible times, i.e. upper bounds on the true minimum) are
33.0, 16.5, 12.0, 8.2, 7.1, 6.9, 7.3 for $r = 0.1, 0.2, 0.3, 0.5, 0.8, 1.0, 1.5$.

The law. Two regimes: for small r the time is the two-level speed limit $\pi/\Omega_{\max}$ (the optimum is only $1.05\times$ that at $r = 0.1$ and 0.2); for large r it flattens at a leakage floor of about 1.1 periods of $2\pi/|\delta|$. Equation (1) joins them with a max. The paper says plainly that this is an empirical fit to a coarse grid, not a theorem; at $r = 0.5$ the max gives 6.91 against the table's 8.2 (the fit misses $r = 0.3$–0.5 by 13–16%). The 2D lab below draws all of it in nanoseconds for the $|\delta|$ you pick.

What is new and what is not. The numbers come from `grape_1q.py`, linked below from the companion package but not re-run for this blueprint (it takes minutes to hours); the lab uses the seven table values as data. For any number of levels the paper proves $T \geq \pi/\Omega_{\max}$ (next node, not formalized), and $T \geq 2\arcsin(1-e)/\Omega_{\max}$ in the trace metric, so for $r \leq 0.2$ the times found are 6% above the bound. The entries for $r \geq 0.8$ are 4-level values and are not converged in the number of levels (see the truncation node). The paper lists the limits: a coarse time grid ($T_{\min}$ known within one step) and no noise or line distortion. Joining the two branches in a certified bound is stated as an open question, not a result.

Evidence & scope

- **Program:** grape_1q.py: 4-level Duffing transmon in units |δ| = 1, piecewise-constant complex drive |Ω| ≤ r|δ|, error < 10^{-4} including leakage; Table V: r = 0.1, 0.2, 0.3, 0.5, 0.8, 1.0, 1.5 give T_found = 33.0, 16.5, 12.0, 8.2, 7.1, 6.9, 7.3 (not re-run here: minutes to hours; the lab uses the table). `grape_1q.py`
- **Program:** the lab's own controls: r = 0.1 gives T|δ| = 32.99, the seven rows of Table V are reproduced, the ratio to π/Ω is 1.05 and 1.30 at r = 0.1 and 0.5, and a claim that the law equals the table at r = 0.5 is rejected (lab/prueba_labs.py). `lab/prueba_labs.py`

Proposition §VII-A. (Multilevel lower bound)

> *A lower bound holds for any number of levels. With* $H(t) = H_{\mathrm{diag}}(t) + \frac{1}{2}(u\,b^\dagger + u^* b)$, $|u| \le \Omega_{\max}$, *and* $p_0 = |a_0|^2$ *the population of* $|0\rangle$, *the only way out of* $|0\rangle$ *is through* $|1\rangle$, *so* $|\dot p_0| \le |u|\,|a_0||a_1| \le |u|\sqrt{p_0(1-p_0)}$ *and the angle* $\theta = \arccos\sqrt{p_0}$ *obeys* $|\dot\theta| \le |u|/2$. *An X gate takes* θ *from* 0 *to* $\pi/2$, *hence* $T \ge \pi/\Omega_{\max}$ *for any truncation; with a transfer fidelity* $|\langle 1|\psi_T\rangle|^2 \ge 1-\varepsilon$, $T \ge 2\arccos\sqrt{\varepsilon}/\Omega_{\max}$. *(This short proof is ours and not formalized; the two-level case is* `QSL.Gate.two_arccos_sqrt_le_time_approx`.*) In the trace metric of Table V the same argument bounds* $|U_{10}|, |U_{01}| \le \sin(\Omega_{\max}T/2)$ *for* $T \le \pi/\Omega_{\max}$, *so an error* e *needs* $T \ge 2\arcsin(1-e)/\Omega_{\max}$; *for* $r \le 0.2$ *the times found are* 6% *above it (*$e = 10^{-4}$*), so there the minimum is known to* 6%; *at IBM's cap (*$\Omega_{\max} = 0.1|\delta|$, $|\delta| = 2.07$ *rad/ns) the archived* 35.6 *ns lies between* 2.2 *and* 2.4 *times the true minimum (bound* 15.0 *ns for that tolerance, pulse found* 15.9 *ns).*

> 🆕 **Classical and new:** a short proof of the paper's own, not formalized in Lean; the two-level case with tolerance is QSL.Gate.two_arccos_sqrt_le_time_approx (commit 42f307b). The paper's «within 5 %» of the earlier draft is now 6 % in the trace metric.

↳ Used in: §VII-A (p. 32)

Evidence & scope

- **By hand:** proof in the paper: with p₀ = |a₀|² the population of |0⟩, the only way out of |0⟩ is through |1⟩, so |ṗ₀| ≤ |u|√(p₀(1−p₀)) and θ = arccos √p₀ obeys |θ̇| ≤ |u|/2; an X gate takes θ from 0 to π/2, hence T ≥ π/Ω_max for any truncation; with transfer fidelity ≥ 1−ε, T ≥ 2 arccos √ε /Ω_max; in the trace metric the same argument bounds |U₁₀|, |U₀₁| ≤ sin(Ω_max T/2) for T ≤ π/Ω_max, so an error e needs T ≥ 2 arcsin(1−e)/Ω_max
- **Program:** r = 0.1: bound 31.133081/|δ|, found 33.0, ratio 1.059966; r = 0.2: bound 15.566541/|δ|, found 16.5, ratio 1.059966; at IBM's cap (|δ| = 2.07) bound 15.040 ns, found 15.942 ns, so 35.6 ns is 2.23–2.37 × the minimum; falsifier threshold (2/π) arcsin(1−10^{-4}) = 0.990997. `tercera_revision.py`, `tercera_revision_salida.txt`

Observation §VII-A. (What IBM could reach)

> For IBM's typical $|\delta| \approx 2.07$ rad/ns, the fastest pulse found at IBM's own cap $r = 0.1$ takes ≈ 16 ns, against 35.6 ns for Falcon single-qubit gates (e.g. `jakarta`): a factor ≈ 2.2 at that cap, which is slightly above the calibrated peak drive (median 0.091). At $r = 1$ the 4-level search finds $T = 6.9/|\delta|$ (3.3 ns), but the truncation matters at this drive (next node). At IBM's cap the 4-level pulse is unaffected by more levels ($\eta_4 = 0.01$). As an independent point of comparison, a 4.16 ns single-qubit pulse on a

superconducting transmon, with 99.76% fidelity and 0.044% leakage, has been demonstrated by optimal control (Werninghaus et al.); its elementary pulses are $\pm X/2$ and $\pm Y/2$ characterized by Clifford-based protocols, a different task from our simulated X gate, so it gives context, not validation. Limits: the time grid is coarse, the search is local, and there is no noise or line distortion in the model.

> 🆕 **Classical and new:** the paper cites a 4.16 ns single-qubit pulse demonstrated by optimal control on a superconducting transmon (Werninghaus et al., 99.76 % fidelity, 0.044 % leakage) as context, not validation: its elementary pulses are ±X/2 and ±Y/2 characterized by Clifford-based protocols, a different task from our simulated X gate. Limits: the time grid is coarse, the search is local, and there is no noise or line distortion in the model.

↳ Used in: §VII-A (p. 31), §XI (p. 45)

🛡️ **Evidence & scope**

- **Program:** 32.99/2.07 = 15.94 ns at r = 0.1; 6.91/2.07 = 3.34 ns for the 4-level value at r = 1; 35.6/15.94 = 2.2 (the factor at the same cap); the table entries for r ≥ 0.8 are 4-level values and are not converged in the number of levels (next node). `lab/prueba_labs.py`

Observation §VII-A. (Truncation)

At $r = 1$ the 4-level search finds $T = 6.9/|\delta|$ (3.3 ns), but the truncation matters at this drive: the 4-level pulse at $T = 7.0/|\delta|$ keeps $1 - F = 7 \cdot 10^{-9}$ in 4 levels but degrades to $5.5\text{–}7.2 \cdot 10^{-4}$ with 5 to 12, and the Duhamel bound on the distance to a larger model, $\eta_4 = \int |u| \, \|\langle 3|UP\| \, dt = 0.55$, does not certify it. A pulse optimized with 6 levels at the same time reaches $1 - F = 4.0 \cdot 10^{-6}$ there and $3.6 \cdot 10^{-6}$ with 8 and 12 levels, so a gate of ≈ 3.4 ns ($T = 7.0/|\delta|$), more than ten times shorter than IBM's, survives in the converged model; it is a numerical witness in an idealized model, not a physical floor. The table entries for $r \geq 0.8$ are 4-level values and are not converged in the number of levels.

> 🆕 **Classical and new:** a numerical witness in an idealized model, not a physical floor. The table entries for r ≥ 0.8 are 4-level values and are not converged in the number of levels; the converged fastest gate, ≈ 3.4 ns, comes from a 6-level search checked up to 12 levels. The Duhamel bound $\eta_4 = 0.55$ does not certify the 4-level pulse.

↳ Used in: §VII-A (p. 32)

🛡️ **Evidence & scope**

- **Program:** the saved 4-level pulse (r = 1, T = 7.0/|δ|) re-evaluated unchanged with L levels: $1-F = 7.19 \cdot 10^{-9}$ (L = 4), $5.50 \cdot 10^{-4}$ (5), $6.88 \cdot 10^{-4}$ (6), $7.20 \cdot 10^{-4}$ (8, 12); Duhamel $\eta_4 = 0.551$. The saved 6-level pulse at the same time: $3.98 \cdot 10^{-6}$ (6 levels), $3.61 \cdot 10^{-6}$ (8 and 12 levels). The paper quotes $7 \cdot 10^{-9}$, $5.5\text{–}7.2 \cdot 10^{-4}$, $\eta_4 = 0.55$, $4.0 \cdot 10^{-6}$ and $3.6 \cdot 10^{-6}$: all agree (re-evaluated for this blueprint; no reoptimization). `truncacion_reevaluar.py`, `truncacion_reevaluar_salida.txt`
- **Program:** archived output of the 6-level optimizations at r = 1.0 (T = 7.0, 8.2, 10.0, 12.0) and at r = 0.351 (T = 9.4, 10.0, 10.8, 11.6), each re-propagated with 4, 5, 6, 8, 12 levels (long optimization; not re-run); the saved pulses are in the package.

`truncacion_1q.py`, `truncacion_L6_salida.txt`, `rfull_salida.txt`, `pulso_x_r1.0_T7.0_L4.npy`, `pulso_x_r1.0_T7.0_L6.npy`

Observation §VII-A. (The full amplitude of IBM)

The drive fraction $r = 1$ is not available in the analysed snapshots: the full amplitude $|D| = 1$ corresponds to $r_{\text{full}} = \texttt{omegad}/|\delta|$, median 0.351 (5th–95th percentile $[0.24, 0.61]$). At $r = 0.351$ a 6-level search finds an X at $T = 10.0/|\delta| = 4.8$ ns with $1 - F = 2.8 \cdot 10^{-8}$, unchanged with 12 levels, while the bound above gives $T \geq 8.87/|\delta| = 4.3$ ns; so within the amplitude budget of the median device a 35.6 ns Falcon gate is 7.4 times the fastest gate found and at most 8.3 times the bound, i.e. 7.4–8.3 times the minimum of the model, and the median gate (11.3 τ_{QSL}, with $\tau_{\text{QSL}} = 8.95/|\delta|$ at $r = 0.351$) is 10–11 times.

> 🆕 **Classical and new:** a model prediction within IBM's own amplitude budget: the 3.4 ns gate needs about three times that amplitude (r = 1 against r_full = 0.351). The drive fraction r = 1 is not available in the analysed snapshots.

↳ Not used by another result of the paper.

🛡️ **Evidence & scope**

- **Program:** r_full = omegad/|δ| over 1646 qubits: median 0.351, 5–95
- **Program:** the 6-level pulse at r = 0.351, T = 10.0/|δ| (4.8 ns): $1-F = 2.83\cdot10^{-8}$ at 6, 8 and 12 levels ($3.15\cdot10^{-8}$ at 5; $1.83\cdot10^{-7}$ at 4); the bound $T \geq 8.87/|\delta| = 4.3$ ns; 35.6/4.8 = 7.4 and 35.6/4.3 = 8.3 for the Falcon gate; τ_QSL = 8.95/|δ| at r = 0.351, so the minimum lies between 0.99 τ_QSL (bound) and 1.12 τ_QSL (found) and the median gate (11.3 τ_QSL) is 10–11 times it. `truncacion_reevaluar.py`, `truncacion_reevaluar_salida.txt`, `rfull_salida.txt`, `pulso_x_r0.351_T10.0_L6.npy`

Observation §VII-B. (Cross-resonance CNOT by optimal control)

We optimized a CNOT on two 3-level transmons with a cross-resonance drive bounded by $|\Omega_{\text{CR}}| \leq r\Delta$ at IBM median parameters ($J/\Delta = 0.0176$, $\delta/\Delta = -2.7$, 10 segments). In units of the exchange time $\pi/(2J)$ the shortest feasible times found (error $<10^{-3}$, upper bounds on the minimum) are 3.27 ($r = 0.25$), 1.82 ($r = 0.45$, IBM's calibrated fraction) and 1.50 ($r = 0.7$). Since a larger cap admits every pulse of a smaller one, 1.50 is also feasible for every $r \geq 0.7$; independent searches at $r = 1.0$ and 1.5 returned longer times, a failure of the local search, which we confirmed by re-expressing the $r = 0.7$ pulse under the larger caps (identical error) and continuing from it to shorter times, which did not reach 10^{-3} down to 1.2. The searches use coarse time grids and few random starts; a repeated run at $r = 0.7$, $T = 1.50$ reached only $1.2 \cdot 10^{-3}$. What the data support is that at IBM's fraction the fastest CNOT found is ≈ 1.2 times slower than at $r = 0.7$; they do not establish an optimum in r.

> 🆕 **Classical and new:** feasible times found, upper bounds on the minimum, on coarse grids with few random starts; they do not establish an optimum in r. The paper does not claim an interior optimum.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** grape_cr.py: CNOT on two 3-level transmons, $|\Omega_{CR}| \le r\Delta$, IBM median parameters ($J/\Delta = 0.0176$, $\delta/\Delta = -2.7$, 10 segments); shortest times found (error $< 10^{-3}$), in units of $\pi/(2J)$: 3.27 ($r = 0.25$), 1.82 ($r = 0.45$), 1.50 ($r = 0.7$) (quoted from the paper; not re-run: long). `grape_cr.py`
- **Program:** monotonicity check: the $r = 0.7$ pulse re-expressed under caps $r = 1.0$ and 1.5 has the identical error ($1.20 \cdot 10^{-3}$), and continuing from it to shorter times does not reach 10^{-3} down to 1.2 (errors $1.7 \cdot 10^{-3}$, $1.0 \cdot 10^{-2}$, $6.3 \cdot 10^{-3}$ at 1.4, 1.3, 1.2 for $r = 1.0$; $1.1 \cdot 10^{-2}$, $1.2 \cdot 10^{-2}$, $5.0 \cdot 10^{-3}$ for $r = 1.5$); independent searches at $r = 1.0$ and 1.5 had returned longer times, a failure of the local search. `grape_cr_continuacion.py`, `cr_continuacion_salida.txt`

Observation §VII-B. (CZ with the second level)

Two 3-level transmons with a fixed exchange coupling $J(b_1^\dagger b_2 + \text{h. c.})$ and fast local control (any Hermitian A_k, B_k on each transmon, 8 piecewise-constant segments, exact gradients). The target is CZ on the qubit subspace with no leakage; the action outside it is free. Every solution is recomputed independently in float64. As a control, the qubit-only search reproduces the known time $\pi/(2J)$ (a gate at $\pi/(2J)$, none at $0.89\,\pi/(2J)$). Write T_* for the minimal time of an exact CZ with ideal (unbounded, instantaneous) local control, and $T_*(e)$ for the minimal time of a CZ with error at most e in the trace metric used throughout. With local controls of at most $10J$ during the gate (plus instantaneous local unitaries before and after), a CZ is realized to error $\le 2 \cdot 10^{-8}$ (leakage $\le 4 \cdot 10^{-8}$) at $T = 0.55/J$, i.e. at 0.35 of the qubit-only time and 5% above the construction. At $T = 0.45/J$ and $0.50/J$, relaxing the cap to $20J$, $40J$ or none did not improve the solutions found ($\approx 5 \cdot 10^{-3}$ and $\approx 4 \cdot 10^{-4}$); this does not identify what limits them, and none was found at $0.50/J$. For the same task and metric, $0.4510/J \le T_*(2 \cdot 10^{-8}) \le 0.55/J$; the bound also shows that no protocol reaches $e \le 2 \cdot 10^{-8}$ at $T = 0.45/J$ (it needs $e \ge 1.9 \cdot 10^{-6}$ there), consistent with the $5 \cdot 10^{-3}$ found. The solutions populate the second level: up to 60% of the population of a computational input leaves the qubit subspace during the gate. Each code state of the construction has population $1/3$ in $|2\rangle$, so each encoded basis input has $5/9 \approx 56\%$ outside the qubit subspace (superpositions range from 0 to $8/9$); together with the $\sqrt{2}$-enhanced elements $\langle 20|H_c|11\rangle$, $\langle 02|H_c|11\rangle$ identified for CNOT and iSWAP, this suggests a mechanism for our solutions that we have not isolated. For the qutrit CZ, $CZ_3|k,l\rangle = e^{2\pi i k l/3}|k,l\rangle$, with a capacitive coupling and drive cap $40g$, the optimal-control time reported by Basyildiz, Gong and Ashhab is close to the qubit limit. That is a different gate: on the qubit subspace it is $\mathrm{diag}(1,1,1,e^{2\pi i/3})$, not $\mathrm{diag}(1,1,1,-1)$, and coupling and controls differ, so the two times are not comparable.

Lean · ours, in part **show**

the entangling-time bound (J = ‖H_c‖ = 2J for d = 3, giving π/(8J)), its robust form with tolerance ε and the 97/32 lemma of the refined bound (QSL.Exchange.refined_cz_lemma; see r:czrefined); the CZ solutions themselves are numerical witnesses, not Lean statements

- `QSL.Qudit.pi_div_four_sub_le_entangling_time_approx`
- `QSL.Qudit.arccos_le_entangling_time`
- `QSL.Exchange.refined_cz_lemma`

✅ Checked when this page was built: ApproxSpeedLimits.lean, QuditEntangling.lean, RefinedCZBound.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — pi_div_four_sub_le_entangling_time_approx, arccos_le_entangling_time, refined_cz_lemma: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean ApproxSpeedLimits.lean QuditEntangling.lean RefinedCZBound.lean` (SHA-256 of the compiled file: `a1ac6d198aa29365…`; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** feasible times found, not proven minima: the shortest accepted solution is at 0.55/J, none was found at 0.50/J (a time at which nothing is found proves nothing). For the same task and metric 0.4510/J ≤ T_*(2·10⁻⁸) ≤ 0.55/J (refined bound, r:czrefined); the bound also shows that no protocol reaches e ≤ 2·10⁻⁸ at T = 0.45/J (it needs e ≥ 1.9·10⁻⁶ there), consistent with the 5·10⁻³ found. The extra level is used: up to 60 % of the population of a computational input leaves the qubit subspace during the gate (each encoded basis input of the construction has 5/9 ≈ 56 % outside; superpositions range from 0 to 8/9); the mechanism is not isolated in our solutions. The qutrit CZ of Basyildiz–Gong–Ashhab is a different gate and their time is not comparable.

↳ Used in: §XI (p. 45)

💡 In plain words (commentary, not part of the paper)

What it says. A CZ gate needs the two qubits to interact. If the two transmons are treated as plain qubits with a fixed exchange coupling J and free, instantaneous local control, the known shortest time is $\pi/(2J)$. If the second level $|2\rangle$ of each transmon is allowed to take part, numerical optimal control finds a CZ with error $2 \cdot 10^{-8}$ in $T = 0.55/J$, with local controls of at most $10J$ during the gate: 0.35 of the qubit-only time.

What is certified and what is not. The entangling-time theorem gives a lower bound for this setting, $\pi/(8J) = 0.393/J$ (the interaction norm is $2J$ for three levels). An explicit construction (fast local twirling, then a code in the extreme eigenvectors of Q) gives a CZ in $\pi/(6J) = 0.5236/J$, so $\pi/(8J) \le T_* \le \pi/(6J)$. A sharper argument (the gate must act coherently on all logical inputs; its $97/32$ lemma is certified in Lean, the time argument is not) raises the lower bound to $0.4511/J$, so $0.4511/J \le T_* \le \pi/(6J)$, and for a gate with error $e = 2 \cdot 10^{-8}$ it gives $0.4510/J \le T_*(2 \cdot 10^{-8}) \le 0.55/J$ (preparing the single state $CZ|{+}{+}\rangle$ is easier and is attained exactly; it is a valid but weak bound for the gate). The solutions are feasible times found, not proven minima; at $0.50/J$ none was found, which proves nothing.

Where the numbers come from. Every saved pulse (19 in the companion package) is re-evaluated independently in float64 by `verificar_cz.py`; `epsilon_cz.py` gives the state tolerance and the robust bound, `tercera_revision.py` the Choi-state bound, `cruces_area_cz.py` the construction, and `poblacion_cz.py` shows that up to 60% of a computational input leaves the qubit subspace (an encoded basis input of the construction has $5/9 \approx 56\%$ outside): the extra level is used, not merely allowed. The control set (arbitrary local Hermitian generators and instantaneous local unitaries) is much wider than a microwave drive on a Duffing transmon.

🛡️ Evidence & scope

- **Program:** grape_cz_torch.py: two 3-level transmons, fixed exchange J(b₁†b₂ + h.c.), any Hermitian local generator on each transmon bounded by a_max, 8 piecewise-constant segments, exact gradients; target CZ on the qubit subspace, no leakage. Table VI = best error 1−F against T and a_max (none, 40J, 20J, 10J), its caption giving the qubit-only limit π/(2J) = 1.571/J, the lower bounds π/(8J) = 0.393/J (certified) and 0.4511/J (refined) and the construction π/(6J) = 0.524/J; the control with two levels reproduces the known time π/(2J) (a gate at π/(2J), none at 0.89 π/(2J)). Long runs (PyTorch); the 19 saved pulses are in the package. `grape_cz_torch.py`, `cz_tope_salida.txt`
- **Program:** every saved solution recomputed independently in float64: e.g. T = 0.55/J, a_max = 10J: 1−F = 2.31·10⁻⁸, leakage 3.9·10⁻⁸, max ‖A_k‖ = 10 J; the other cells of the table agree (re-run here: 19 pulses, deterministic). `verificar_cz.py`, `verificar_cz_salida.txt`
- **Program:** state tolerance for the same task at both ends: for |++⟩ the 10J solution has output infidelity ε = 5.2·10⁻⁸, and the robust bound (π/4 − arcsin √ε)/(2J) = 0.39259/J holds for that same tolerance. `epsilon_cz.py`, `epsilon_cz_salida.txt`
- **Program:** population outside the qubit subspace during the T = 0.55/J, 10J gate: up to 60

Observation §VII-B. (An explicit construction of the CZ)

CZ maps $|{+}{+}\rangle$ to a maximally entangled state, so the entangling-time theorem with $\|H_c\| = 2J$ $(d = 3)$ gives $T_* \geq \pi/(8J) = 0.393/J$. An explicit construction gives an upper bound. Writing $H_c = \frac{J}{2}(Q \otimes Q + \Pi \otimes \Pi)$ with $Q = b + b^\dagger$ and $\Pi = i(b^\dagger - b)$, fast conjugations of *one* transmon by $W = V_Q \,\mathrm{diag}(1, e^{2\pi i/3}, e^{4\pi i/3}) V_Q^\dagger$ (V_Q diagonalizing Q) average $\Pi \otimes \Pi$ away and keep $\frac{J}{2} Q \otimes Q$; encoding each qubit in the two extreme eigenvectors of Q (eigenvalues $\pm\sqrt{3}$) gives $\frac{3J}{2} Z \otimes Z$, hence a CZ (up to local phases) in interaction time $\pi/(6J) = 0.5236/J$, in the limit of arbitrarily fast local operations So $\pi/(8J) \leq T_* \leq \pi/(6J)$; the lower bound is sharpened to $0.4511/J$ in the next nodes (the construction is then within 16% of the bound).

Lean · ours, in part **show**

only the lower bound π/(8J) (entangling-time theorem with ‖H_c‖ = 2J, d = 3); the upper bound, the explicit construction, is not formalized

- `QSL.Qudit.arccos_le_entangling_time`

✅ Checked when this page was built: QuditEntangling.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — arccos_le_entangling_time: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean QuditEntangling.lean` (SHA-256 of the compiled file: `f4a5538fe9427bac…`; a source page on this site may show it with comments edited, under its own hash).

NEW **Classical and new:** an upper bound on T_*, in the limit of arbitrarily fast local operations. Not in Lean.

↳ Used in: §VII-B (p. 33), §VII-B (p. 38)

Evidence & scope

- **Program:** H_c = (QQ+PP)/2 checked; the twirl by W (on one transmon) equals QQ/2; eigenvalues of Q: −1.7321, 0, 1.7321; the logical block at T = π/6 is unitary with relative phases/π = (0, 0.5, 0.5, 0), i.e. a CZ up to local Z rotations; p_2 of the extreme eigenvectors 1/3, so each encoded basis input has 0.555556 = 5/9 outside the qubit subspace. `cruces_area_cz.py`, `cruces_area_cz_salida.txt`

Observation §VII-B. (The Choi-state bound for the gate)

For a gate with error e, add a two-level reference to each transmon and start from the product (across AA':BB') of two local maximally entangled states: the overlap of the resulting state with the one produced by CZ is exactly $|\mathrm{Tr}(CZ^\dagger PUP)|/4 = 1 - e$, leakage included, and the CZ state has Schmidt coefficients $(1/\sqrt{2}, 1/\sqrt{2})$ across that cut, while the references leave $\|H_c\| = 2J$ unchanged. The tolerance version of the theorem (with $\varepsilon = 1 - (1-e)^2$) then gives $T_*(e) \geq [\pi/4 - \arccos(1-e)]/(2J)$ (this instantiation with references is ours, not formalized). Preparing the single output $CZ|{+}{+}\rangle$ is a much easier task: local operations map $|{+}{+}\rangle$ to $|12\rangle$, on $\{|12\rangle, |21\rangle\}$ the coupling acts as $2J\sigma_x$, and local operations map the result back, so the state-preparation bound is attained exactly: it is a valid but weak bound for the gate. The norm-based bound is sharpened in the next node: $0.4510/J \leq T_*(2 \cdot 10^{-8}) \leq 0.55/J$.

Lean · ours, in part **show**

the tolerance theorem (ε the infidelity of one output state); its instantiation with two-level references on each transmon, which turns it into a bound on the gate in the trace metric, is ours and not formalized

- `QSL.Qudit.pi_div_four_sub_le_entangling_time_approx`

✅ Checked when this page was built: ApproxSpeedLimits.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — pi_div_four_sub_le_entangling_time_approx: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean ApproxSpeedLimits.lean` (SHA-256 of the compiled file: `a1ac6d198aa29365…`; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** for a gate with trace error e the overlap of the extended state with the one produced by CZ is exactly |Tr(CZ†PUP)|/4 = 1−e, leakage included, with ε = 1−(1−e)²; preparing the single output CZ|++⟩ is a much easier task: local operations map |++⟩ to |12⟩, on {|12⟩, |21⟩} the coupling acts as 2Jσₓ, so the state-preparation bound is attained exactly: it is a valid but weak bound for the gate.

↳ Used in: §VII-B (p. 33), §VII-B (p. 36)

🛡️ **Evidence & scope**

- **Program:** exact Bell preparation: ⟨21|H_c|12⟩ = 2.0 and H_c|12⟩ stays in span12, 21; at T = 0.392585/J the Bell infidelity is $5.2 \cdot 10^{-8}$. Choi bound: T_CZ(e = $2 \cdot 10^{-8}$) ≥ (π/4 − arccos(1−e))/2 = 0.392599081699/J; the overlap identity holds. `tercera_revision.py`, `tercera_revision_salida.txt`
- **Program:** the 10J solution has output infidelity ε = $5.2 \cdot 10^{-8}$ on |++⟩ and the robust bound 0.39259/J (the older state-tolerance version). `epsilon_cz.py`, `epsilon_cz_salida.txt`

Observation §VII-B. (The refined CZ bound)

The gate bound itself can be sharpened, because the gate must act coherently on all logical inputs (the lemma is certified in Lean as `QSL.Exchange.refined_cz_lemma`, with its equality case, and checked numerically in `cota_cz_refinada.py`; the time argument built on it is not formalized). Let $L(t)$ apply the same local controls as $U(t)$ but without H_c, and $z(t) = U(t)^\dagger L(t)|\Omega_0\rangle$ with $|\Omega_0\rangle$ the reference-extended product state above. Then $\dot z = iU^\dagger H_c L|\Omega_0\rangle$, whose Fubini–Study speed is the spread of H_c in $L(t)|\Omega_0\rangle$; that state has physical reduced state $(P_A \otimes P_B)/4$ with P_A, P_B rank-2 projectors, and for any such projectors $\frac{1}{4}\mathrm{Tr}[(P_A \otimes P_B)H_c^2] \le \frac{97}{32}J^2$ (equality at $P = \mathbb{1} - |v\rangle\langle v|$, $|v\rangle = \sqrt{7/8}|0\rangle + \sqrt{1/8}|2\rangle$). Since $L(T)|\Omega_0\rangle$ is a product across $AA'{:}BB'$ and the CZ state is at Fubini–Study distance at least $\pi/4$ from every product, $T_*(e) \ge [\pi/4 - \arccos(1-e)]/(J\sqrt{97/32})$, i.e. $T_* \ge 0.4511/J$ and $T_*(2\cdot 10^{-8}) \ge 0.4510/J$, 15% above the norm bound. With ideal local control, $0.4511/J \le T_* \le \pi/(6J) = 0.5236/J$ (the upper value as a limit of finite protocols); the construction is within 16% of the bound. Preparing one Bell state and implementing the whole CZ therefore have different minimal times in this model.

Lean · ours, in part **show**

the lemma (1/4)·Tr[(P_A⊗P_B)H_c²] ≤ (97/32)J² for rank-2 projectors on d = 3, and its equality case at P = 1 − |v⟩⟨v|, |v⟩ = √(7/8)|0⟩ + √(1/8)|2⟩; the time argument built on it (the comparator trajectory L(t), the Fubini–Study speed, the distance π/4 from the product states) is not formalized

- `QSL.Exchange.refined_cz_lemma`
- `QSL.Exchange.refined_cz_equality`

✅ Checked when this page was built: RefinedCZBound.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — refined_cz_lemma, refined_cz_equality: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean RefinedCZBound.lean` (SHA-256 of the compiled file: `0430d3f31d858be6…`; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** the lemma is certified in Lean and checked numerically by us, the time argument is not formalized. The bound is 15 % above the norm bound π/(8J) = 0.393/J and the construction π/(6J) is within 16 % of it. Preparing one Bell state and implementing the whole CZ therefore have different minimal times in this model. No protocol reaches e ≤ 2·10⁻⁸ at T = 0.45/J (it needs e ≥ 1.9·10⁻⁶ there), consistent with the 5·10⁻³ found; the upper value π/(6J) is a limit of finite protocols.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** the lemma: max Tr[(P_A⊗P_B)H_c²]/4 over 300 random starts of rank-2 projectors = 3.0312500000 = 97/32; at P = 1 − |v⟩⟨v|, |v⟩ = √(7/8)|0⟩ + √(1/8)|2⟩: ⟨H_c²⟩ = 3.0312500000 and ⟨H_c⟩ = 0; 2·10⁵ random projector pairs give a maximum of 2.992860 ≤ 97/32. `cota_cz_refinada.py`, `cota_cz_refinada_salida.txt`
- **Program:** the resulting time bound J·T ≥ (π/4 − arccos(1−e))/√(97/32): 0.4511064 for e = 0 (against 0.3926991 before), 0.4509915 for e = 2·10⁻⁸, 0.4429836 for e = 10⁻⁴; the smallest possible error at T = 0.45/J is 1.855·10⁻⁶ (paper: 1.9·10⁻⁶; compatible with the

5.0·10⁻³ found), and at 0.50/J and 0.55/J the bound allows error 0 (compatible with 3.7·10⁻⁴ and 2·10⁻⁸); π/6 against the bound: 1.1607 (paper: within 16

- **By hand:** the argument in the paper: L(t) applies the same local controls as U(t) without H_c, z(t) = U(t)†L(t)|Ω₀⟩ with |Ω₀⟩ the reference-extended product state; ż = iU†H_c L|Ω₀⟩, whose Fubini–Study speed is the spread of H_c in L(t)|Ω₀⟩; that state has physical reduced state (P_A⊗P_B)/4 with P_A, P_B rank-2 projectors, so the speed is at most J√(97/32); L(T)|Ω₀⟩ is a product across AA′:BB′ and the CZ state is at Fubini–Study distance at least π/4 from every product, hence T_*(e) ≥ [π/4 − arccos(1−e)]/(J√(97/32)). Quoted from the paper; not formalized

Observation §VII-B. (Larger local dimension)

The same construction works for d-level transmons: encoding each qubit in the two extreme eigenvectors of $Q_d = b + b^\dagger$ (eigenvalues $\pm\lambda_d$, with $\lambda_d/\sqrt{2}$ the largest zero of the Hermite polynomial H_d) gives a CZ in interaction time $\pi/(2J\lambda_d^2)$, while the entangling-time theorem gives $\pi/(4J\kappa_d)$ with $\kappa_d = \|b^\dagger \otimes b + b \otimes b^\dagger\|$. The ratio of the two bounds is $2\kappa_d/\lambda_d^2$: 2, 1.33, 1.27, 1.20, 1.09, 1.05 for $d = 2, 3, 4, 5, 10, 20$, and it tends to 1. This is certified in Lean: $\lambda_d^2/2 \le \kappa_d \le 2(d-1)$ (`QSL.Exchange.exchange_norm_ge`, `QSL.Exchange.exchange_norm_le`), $\lambda_d \ge 2(1 - 1/m)\sqrt{d-m}$ for the largest eigenvalue (`QSL.Exchange.lamMax_ge`), and $2\kappa_d/\lambda_d^2 \to 1$ (`QSL.Exchange.ratio_tendsto_one`); the rate is $O(d^{-2/3})$ by the asymptotics of the largest Hermite zero (DLMF 18.16.17, not formalized). At a fixed interaction norm $\|H_c\| = J\kappa_d$, the CZ time of the construction therefore approaches the Bell-preparation bound: the entangling-time bound becomes tight for the whole gate as $d \to \infty$, with ideal local control. The averaging step is an instance of the no-overhead simulation framework of Wocjan, Rötteler, Janzing and Beth (2002); the quadrature code and the asymptotic tightness were not found in the sources we read.

Lean · ours, in part **show**

λ_d²/2 ≤ κ_d ≤ 2(d−1) (exchange_norm_ge, exchange_norm_le), λ_d ≥ 2(1 − 1/m)√(d − m) for the largest eigenvalue (lamMax_ge) and 2κ_d/λ_d² → 1 (ratio_tendsto_one), in Formalizations/Physics/QuantumMechanics/SpeedLimit/ExchangeQuadrature.lean; the rate O(d^(−2/3)) (asymptotics of the largest Hermite zero, DLMF 18.16.17), the d-level CZ construction itself and the table of ratios are not formalized

- `QSL.Exchange.exchange_norm_ge`
- `QSL.Exchange.exchange_norm_le`
- `QSL.Exchange.lamMax_ge`
- `QSL.Exchange.ratio_tendsto_one`

✅ Checked when this page was built: ExchangeQuadrature.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — exchange_norm_ge, exchange_norm_le, lamMax_ge, ratio_tendsto_one: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean ExchangeQuadrature.lean` (SHA-256 of the compiled file: `8326ffdc315391ee…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** the averaging step is an instance of the no-overhead simulation framework of Wocjan, Rötteler, Janzing and Beth (2002); the quadrature code and the asymptotic tightness were not found in the sources read. With ideal local control, at a fixed

interaction norm ‖H_c‖ = Jκ_d the CZ time of the construction approaches the Bell-preparation bound: the entangling-time bound becomes tight for the whole gate as d → ∞.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** the table of bounds for d = 2, 3, 4, 5, 10, 20: lower π/(4κ_d) = 0.785398, 0.392699, 0.226725, 0.160319, 0.060939, 0.025699; upper π/(2λ_d²) = 1.570796, 0.523599, 0.288246, 0.192446, 0.066519, 0.027059; ratio 2.0000, 1.3333, 1.2713, 1.2004, 1.0916, 1.0529 (paper: 2, 1.33, 1.27, 1.20, 1.09, 1.05). `cruces_area_cz.py`, `cruces_area_cz_salida.txt`
- **By hand:** the rate O(d^(−2/3)) follows from the asymptotics of the largest zero of the Hermite polynomial H_d (DLMF 18.16.17); quoted from the paper, not formalized

Evidence & scope

- **Program:** for H = XX+YY+ZZ and fast local control, the smallest largest-Schmidt-coefficient found from a product state is λ(T) = cos(2T) to 10^{-13} without ancillas and to 10^{-8} with one ancilla qubit per side; the same identity λ(T) = cos(wT) holds for Ising ZZ and XY couplings, with and without ancillas. `grape_bell_torch.py`, `bell_anchura_salida.txt`, `grape_anchura.py`, `anchura_u4d_salida.txt`
- **By hand:** analytic witness: since H = 2·SWAP − 1, e^−iHT|01⟩ = e^iT[cos(2T)|01⟩ − i sin(2T)|10⟩], whose largest Schmidt coefficient is cos(2T) for T ≤ π/8; with the certified bound λ(T) ≥ cos(2T) the minimum is exactly cos(2T) up to the Bell time, and the numerics are checks of this

8 Application: A Chip-to-Chip State Transfer

Remark §VIII. (The link and the model)

The same diagnostic applies to operations other than gates. Sundaresan *et al.* transfer a qubit state between two 120-qubit processors through an 80 cm aluminium coaxial link (an "L-coupler"), relaying the excitation through one standing mode of the cable. They report the maximal qubit–mode coupling, $g/2\pi = 8$–9 MHz (their Fig. 1f), the transfer duration T_g and the fidelity of each of four links (their Extended Data Table 2), and the coherence times of qubits and cable modes (Extended Data Tables 1 and 2). The coupling waveforms are not published. *Model.* In the single-excitation subspace, with the rotating-wave approximation and one resonant mode, the relay is a three-level problem on $|A\rangle$ (first qubit), $|C\rangle$ (cable mode), $|B\rangle$ (second qubit): $H(t) = g_1(t)\,(|A\rangle\langle C| + \text{h. c.}) + g_2(t)\,(|C\rangle\langle B| + \text{h. c.})$ with complex $g_i(t)$, $|g_i(t)| \le g$ and $g = 2\pi \times (g/2\pi)$.

NEW **Classical and new:** data of Sundaresan et al. (IBM), arXiv:2610.29986: Fig. 1f (g/2π), Extended Data Tables 1 and 2 (T_g, F, coherence times, read in the image of the tables); the coupling waveforms are not published.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** the published values of the four links, each with its table, as a data file read by every script of the section; diag_relevo.py prints the table of all four links from it (T_g, T_g/T_min, measured error, coherence). `relevo/diag_relevo.py`, `relevo/datos/ibm_2610.29986.toml`

Observation §VIII. (Certified floor)

Since $\|H(t)\| = (|g_1|^2 + |g_2|^2)^{1/2} \le \sqrt{2}\,g$ and the transfer $|A\rangle \to |B\rangle$ needs a Fubini–Study angle $\pi/2$, the bound `QSL.Gate.fsAngle_le_time_halfWidth` of Section II (with shift $c = 0$; it holds in any dimension) gives $T \ge \pi/(2\sqrt{2}\,g)$ for piecewise-constant couplings of any phase. It is certified and not tight.

Lean · ours, in part show

the angle bound fsAngle ≤ (Ω/2)·T for piecewise-constant Hamiltonians in any dimension, applied with shift c = 0 and Ω/2 = √2 g; the norm ‖H‖ = (|g₁|² + |g₂|²)^{1/2} and the angle π/2 of the transfer are the paper's one-line application

- `QSL.Gate.fsAngle_le_time_halfWidth`

✅ Checked when this page was built: GateEigenphases.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — fsAngle_le_time_halfWidth: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean GateEigenphases.lean` (SHA-256 of the compiled file: `0a989bcfb2879f6f…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** standard (shifted Mandelstam–Tamm); the paper says it is certified and not tight (half the optimum).

↳ Used in: §VIII (p. 40)

Observation §VIII. ✅ Isabelle **(Exact optimum)**

For this model the minimal time is $\pi/(\sqrt{2}\,g)$, reached with both couplings constant and at their maximum: the isotropic case of Theorem 1 of Boscain, Chambrion and Charlot. The lower half, $gT \ge \pi/\sqrt{2}$, is machine-checked in Isabelle/HOL (`Relay_Minimal_Time.relay_minimal_time`, entry `Quantum_Speed_Limits` submitted to the Archive of Formal Proofs) for the amplitudes $x = |a_A|$, $z = |a_B|$, which satisfy $|\dot{x}|, |\dot{z}| \le g\sqrt{1 - x^2 - z^2}$ away from a countable set of times; the proof uses an explicit calibration function, and the reduction from the Schrödinger equation to these inequalities is in the text, not in Isabelle. We also reproduced the optimum numerically with piecewise-constant controls, real and complex: no control found reaches full transfer for $gT < \pi/\sqrt{2}$. With these couplings the excitation spends a quarter of the time in the cable ($\int_0^T |c(t)|^2 dt = T/4$). The sequential protocol (two swaps) takes π/g.

Isabelle · ours show

`relay_minimal_time` · `Isabelle/AFP/Quantum_Speed_Limits/Relay_Minimal_Time.thy`, line 519 — π/√2 ≤ G·T for amplitudes x, z ≥ 0 with x² + z² ≤ 1, |x′|, |z′| ≤ G·√(1 − x²

− z²) outside a countable set, x(0) = 1, z(T) = 1 (an explicit calibration function); entry Quantum_Speed_Limits submitted to the AFP, 30-oct-2026

```
theorem relay_minimal_time:
  fixes x z x' z' :: "real \<Rightarrow> real" and G T :: real and S :: "real
set"
  assumes T: "0 \<le> T" and S: "countable S"
    and cx: "continuous_on {0..T} x" and cz: "continuous_on {0..T} z"
    and D: "\<forall>t\<in>{0..T}. 0 \<le> x t \<and> 0 \<le> z t \<and> (x
t)^2 + (z t)^2 \<le> 1"
    and x0: "x 0 = 1" and zT: "z T = 1"
    and der: "\<forall>t\<in>{0<..<T} - S. (x has_real_derivative x' t) (at t)
\<and> (z has_real_derivative z' t) (at t)"
    and bnd: "\<forall>t\<in>{0<..<T} - S. \<bar>x' t\<bar> \<le> G * sqrt (1
- (x t)^2 - (z t)^2) \<and> \<bar>z' t\<bar> \<le> G * sqrt (1 - (x t)^2 - (z
t)^2)"
  shows "pi / sqrt 2 \<le> G * T"
```

🆕 **Classical and new:** the optimum is known: Boscain, Chambrion and Charlot, DCDS-B 5 (2005), Theorem 1, isotropic case. Ours: the Isabelle proof of the lower half by a calibration function; the reduction from the Schrödinger equation to the amplitude inequalities (valid with several modes and detunings when g is the norm of each coupling vector) is in the text, not in Isabelle; attainment is from Boscain et al. and our numerics.

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** no piecewise-constant control, real or complex, reaches full transfer for gT < π/√2; at gT = π/√2 the constant couplings do. `relevo/tiempo_minimo.py`
- **Program:** with constant couplings at the optimum the excitation spends a quarter of the time in the cable (∫|c|² dt = T/4), and the closed evolution transfers with fidelity 1.000000000. `relevo/lindblad_relevo.py`

Remark Table VII. (Table VII and the gaps)

IBM chip-to-chip transfers against the single-mode optimum $\pi/(\sqrt{2}\,g)$, for $g/2\pi = 8$ and 9 MHz (44.2 and 39.3 ns). Q240–Q247: $T_g = 312$ ns, $T_g/T_{\min} = 7.06$–7.94, error 0.71%, coherence at T_g (IBM's model) 0.63%, coherence at $T_{\min}$ 0.13–0.15%. Q241–Q246: 296 ns, 6.70–7.53, 1.13%, 0.87%, 0.23–0.26%. Q242–Q245: 256 ns, 5.79–6.52, 1.01%, 0.61%, 0.16–0.18%. Q243–Q244: 312 ns, 7.06–7.94, 1.14%, 0.94%, 0.22–0.24%. The ratios are ratios to the optimum of the single-mode model with the published maximal couplings; they do not show that the device admits a speed-up by that factor at the same fidelity. The transfers last 5.8–7.9 times the single-mode optimum, 4.1–5.6 times the sequential time and 11.6–15.9 times the certified floor. The 126 ns that the authors give as a target would still be 2.9–3.2 times the optimum.

🆕 **Classical and new:** a computation on IBM's published values; the ratios are to the optimum of the single-mode model with the published maximal couplings and do not show that the device admits that speed-up at the same fidelity.

↳ Used in: §VIII

Evidence & scope

- **Program:** ratios at g/2π = 8, 8.5 and 9 MHz for the four links and the 126 ns target: 7.06–7.94, 6.70–7.53, 5.79–6.52, 7.06–7.94; target 2.85–3.21.

`relevo/cocientes.py`

- **Program:** the lab of this node reproduces the row of Q240–Q247 (T_g/T_min = 7.94 at 9 MHz) and IBM's coherence model (0.625

Evidence & scope

- **Program:** master equation at the minimal time: 0.1289
- **Program:** additive model against the master equation: ratio 1.000 with relaxation only, 1.54 (qubits) and 2.25 (cable) with dephasing only, the exact first-order values 37/24 and 9/4. `relevo/comparar_modelos.py`
- **Program:** not a floor: with lossless cable and lossy qubits the sequential protocol (twice the duration) loses less, 9.26e-5 against 9.82e-5. `relevo/falsar_v198.py`
- **Program:** validation of the simulator against the published Lindblad simulation of an IBM SATD transfer (Heya et al., PRL 135, 200801): 2.04e-3 against 2e-3 (cable loss only) and 6.04e-3 against 6.2e-3 (all losses). `relevo/heya_satd.py`, `relevo/mision_heya_satd.toml`

Remark §VIII. (Caveats)

(i) The cable has many modes (free spectral range ≈ 150 MHz, $g/\mathrm{FSR} \approx 0.05$–$0.06$ by the authors' figures); protocols that use several modes are not covered by the single-mode optimum, and for them the certified floor and the Isabelle bound $\pi/(\sqrt{2}\,G)$ hold with the collective coupling norm G of each end, which is not reported. (ii) g is the maximal coupling; the ratios measure the distance to what that maximum allows, not to the waveforms used. (iii) The reported fidelity and the error model are IBM's; the occupations $\frac{1}{2}$–$\frac{1}{2}$ are IBM's assumption, and for the qubits we use, as IBM does, the shorter coherence time of each pair. (iv) The master equation takes dephasing as Markovian at the rate set by the Ramsey T_2; low-frequency noise would change the dephasing terms.

NEW **Classical and new:** the paper's own scope of Section VIII.

↳ Not used by another result of the paper.

9 What the Certificate Does and Does Not Guarantee

Remark §IX. (What the certificate guarantees)

The certified objects are the inequalities, not the gaps. The theorems of Table I are machine-checked for all inputs satisfying their hypotheses. Each reported gap is a computation on vendor data; its trust rests on the controls of Section III, not on a proof. *The model is IBM's, not a measurement.* Every speed limit is evaluated in IBM's Duffing model with the RWA. If the model misdescribes a device, the gap inherits the error. *`omegad` is derived.* It comes from the π-pulse calibration, and a factor $1/(1-e^{-2})$ separates the area with which `omegad` was derived from Qiskit's normalized pulse; which of the two waveforms reaches the qubit is not in the snapshots. The single-qubit gaps carry this uncertainty, and duration–`omegad` correlations are

partly circular. *Hypotheses of the two-qubit theorem.* The Lean theorem assumes a constant interaction of bounded norm and piecewise-constant local controls, with a product initial state. Applying it to IBM gates assumes that the drives act locally and that the static coupling is the only interaction (no classical crosstalk). The single-qubit time-dependent bound and the ECR corollary are stated in Lean for piecewise-constant controls; a continuous pulse is covered only as a limit of such controls, which is not formalized. The norm $\|H_{\text{int}}\| = j_q$ holds only in the qubit subspace, and j_q itself is a model value repeated across dates. *Bounds that are not in Lean.* The literature gate bound $\pi/(2j_q)$ behind the headline 4.22, its application to real durations (our inference), the multilevel single-qubit bound $T \geq \pi/\Omega_{\max}$ and its trace-metric version (proved in the text), the instantiation with references, the time argument of the refined CZ bound $0.4511/J$ (its $97/32$ lemma is in Lean) and the CZ construction are not formalized (Lean certifies $\pi/(8J)$). *Truncation.* The single-qubit times found with 4 levels at $r \geq 0.8$ are not converged in the number of levels; the converged fastest gate, ≈ 3.4 ns, comes from a 6-level search checked up to 12 levels. *Effective cross-resonance models.* The CR check uses effective Hamiltonians (first order, third order, and a static numerical block diagonalization); none is a time-dependent simulation of the calibrated pulse sequence with echoes and cancellation tones. *Coarse optimal control.* The optimal-control tables come from coarse grids (time steps, 8–10 piecewise-constant segments, finitely many random starts), without noise; the times reported are feasible times found, i.e. upper bounds on the minimum, and a time at which no gate was found proves nothing. *Snapshots, not live data; no hardware validation.* All data are dated snapshots. We have run nothing on hardware: IBM has been withdrawing pulse-level access, so the feasible times found in Section VII are model predictions. Black-box timing on live hardware is nevertheless feasible: gate-time amplification, which reads gate durations from job wall-time, could validate the snapshot durations used here. We have not done so.

🆕 **Classical and new:** the paper's own statement of scope: not in Lean are the literature gate bound π/(2j_q) behind the headline 4.22, its application to real durations (our inference), the multilevel single-qubit bound T ≥ π/Ω_max (proved in the text), the Choi instantiation of the tolerance theorem, the time argument of the refined CZ bound 0.4511/J (its 97/32 lemma is in Lean), the rate O(d^(−2/3)) of the larger-dimension ratio (DLMF 18.16.17) and the CZ construction (Lean certifies π/(8J)); the cross-resonance checks are effective Hamiltonians, not a time-dependent simulation; the optimal-control tables are coarse feasible times found, upper bounds; nothing was run on hardware; the Lean time-dependent and ECR statements cover piecewise-constant controls only.

↳ Not used by another result of the paper.

10 Reproducibility

Remark §X. (Reproducibility)

All scripts and data are in the companion package of the blueprint, https://karlesmarin.github.io/blueprints/qsl-diagnostics/companion/ (Apache 2.0):

`convencion.py` (three-level check of the drive convention), `pulsos_x.py` and `identidad_omegad.py`, `identidad_calibracion.py` (convention from calibrated π pulses), `barrido.py` (single-qubit sweep, `barrido.csv`), `pares.py` (two-qubit pairs, `pares.csv`), `cruce_formulas.py` (CR and peak-Rabi checks), `serie_temporal.py`, `serie_porque.py`, `duracion_error.py` (time series, `serie.csv`), `degradacion.py` and `frecuencia_degradados.py` (stratified controls), `mejora_por_cubit.py` (per-qubit durations), `mediana_por_backend.py` (backend-weighted median, clustered bootstrap), `toronto_sensibilidad.py` (outlier sensitivity), `control_dos_cubits.py` and `cnot_m3.py` (numerical controls of the certified bounds), `grape_1q.py`, `grape_cr.py` and `grape_cr_continuacion.py` (monotonicity check), `grape_cz_torch.py` with its independent check `verificar_cz.py`, `grape_bell_torch.py`, `grape_anchura.py`, `poblacion_cz.py` (optimal control), `fuentes/cruce_mg2020.py` (higher-order CR check), `cruces_area_cz.py` (pulse-area identity, CZ construction) and `truncacion_1q.py` (truncation check), `tercera_revision.py` (gate-metric bounds, r_{full}), `cota_cz_refinada.py` (refined CZ bound). Data: `qiskit-ibm-runtime` 0.51.0 and `qiskit-terra` 0.24.1 fake-backend snapshots (Apache 2.0), copied in `datos/` and `datos_serie/`. The Lean sources are at https://github.com/karlesmarin/formalizations, commit `42f307b`. The calibrated pulses (`qiskit-terra` 0.24.1) are copied in `datos_pulsos/`; a self-check rebuilds `barrido.csv` inside the package and compares it with the archived file. A companion blueprint, https://karlesmarin.github.io/blueprints/qsl-diagnostics/, maps each claim of this paper to its Lean theorem or script. Its laboratories run in the reader's browser on the reader's own values: Figs. 1–4 redrawn live; the fixed-axis equality on the Bloch sphere for any envelope; the certified floors of Section II for a list of the reader's own gates (diagnose your device), which fires the falsifier of Section III on any gate below its floor; and the d-level CZ bounds of Section VII. Each result can be downloaded as CSV or shared as a link with its inputs. For Section VIII (version 2): `relevo/datos/ibm_2610.29986.toml` (the published values, each with its table), `relevo/cocientes.py`, `relevo/diag_relevo.py`, `relevo/lindblad_relevo.py` with its independent check `relevo/lindblad_rk4.py`, `relevo/suelo_lindblad.py`, `relevo/comparar_modelos.py`, `relevo/falsar_v198.py`, `relevo/tiempo_minimo.py` and `relevo/heya_satd.py`, which reproduces the published Lindblad simulation of an IBM SATD transfer as a validation of the simulator. The Isabelle sources of `Quantum_Speed_Limits` are in the same package.

> 🆕 **Classical and new:** the package (scripts, outputs, saved CZ and X pulses, IBM snapshots with their Apache-2.0 licence) is in the «Programs and data» folder of this blueprint; the paper points to it and to the labs by URL; only the archive DOI is still pending in the draft.

↳ Not used by another result of the paper.

🛡️ **Evidence & scope**

- **Program:** the claim-to-script-to-output map of the companion package (its README), the package self-checks (barrido.py rebuilds barrido.csv inside the package;

cruce_mg2020.py reproduces its archived output; cruces_area_cz.py, tercera_revision.py, cota_cz_refinada.py, truncacion_reevaluar.py and mediana_por_backend.py are re-run when the package is built and their outputs checked against the numbers quoted in the paper; the saved CZ and X pulses are re-evaluated deterministically), and the recomputation of the headline numbers from the archived outputs barrido.csv, pares.csv and serie.csv with controls that must fail (lab/comprueba_numeros.py). `README.md`, `lab/comprueba_numeros.py`

11 Conclusion

Remark §XI. (Conclusion)

On public calibration snapshots of pre-Heron IBM processors (mainly Falcon and Eagle), single-qubit gates run at a median $11.3\times$ their two-level speed limit at full drive amplitude, a ratio within about 5% of 11.3 in every year from 2021 to 2025. That denominator is an amplitude IBM does not use: it runs the drive at about a tenth of the anharmonicity, and at that drive its pulses are about 2 times the square-pulse limit (the shape factor 1.87, or 2.16 with Qiskit's peak normalization) and can be shortened by at most 2.18 (nominal peak) or 1.89 (edge-subtracted peak) at the same peak drive; the remaining ≈ 5 is amplitude headroom, a design choice. Optimal control on IBM's model finds ≈ 16 ns X gates at IBM's own amplitude cap and 4.8 ns at its full amplitude (where the minimum is at least 4.3 ns): within the same amplitude budget the model admits gates about 10 times faster than the median gate (7.4–8.3 times for a 35.6 ns Falcon gate); ≈ 3.4 ns needs three times that amplitude. These are feasible times in an idealized model, not physical floors. Two-qubit gates run at a median $4.22\times$ the known qubit-subspace CNOT-class bound $\pi/(2j_q)$; in the snapshots that ratio rose from 2.78 to 4.26 between 2021 and 2025 as gates got longer, not monotonically and coinciding with the change from Falcon CX to Eagle ECR, while the archived couplings did not change. The current Heron generation (CZ ≈ 68–70 ns) is not covered. The inequalities of Table I are machine-checked, including an any-dimension entangling-time bound that is new only as a certified assembly of known pieces. On the model, optimal control finds a cross-resonance CNOT ≈ 1.2 times faster at drive fraction 0.7 than at IBM's 0.45, and, with the second transmon level, a CZ (error $2 \cdot 10^{-8}$) in 0.35 of the qubit-only exchange time with local controls of at most $10J$; both are feasible times found, not proven minima. The open questions are the true minimum between the refined bound $0.4511/J$ and the explicit construction $\pi/(6J)$ (in d-level transmons the gap between the entangling bound and the construction closes as $d \to \infty$, a limit certified in Lean), and a certified bound that joins the speed-limit and leakage branches of Eq. (1), for which we have no result. The same diagnostic applied to IBM's chip-to-chip state transfer (version 2) places it at 5.8–7.9 times the single-mode minimal time $\pi/(\sqrt{2}\,g)$, whose lower bound is machine-checked in Isabelle, with a coherence error at that time four to six times below the measured errors; this is the error of one protocol, not a floor.

🆕 **Classical and new:** the machine-checked entangling-time bound is «new only as a certified assembly of known pieces»; the last open question (a certified bound joining the two branches) is, in the paper, a conjecture only, with no result.

↳ Not used by another result of the paper.

Remark §XI. (Conflict of interest and use of AI assistants)

The author declares no conflict of interest. This work received no funding. *Use of AI tools.* This work was carried out with an AI assistant. The author checked all statements, proofs and computations and takes full responsibility for the content.

🆕 **Classical and new:** version 2 states the use of AI assistants in its own section, not in an acknowledgment.

↳ Not used by another result of the paper.

📊 Tables

Every table of the paper, in its order: click one to unfold it, with where the paper cites it, how its numbers were checked, and its data as CSV. The typeset table is one click away, on its page of the PDF.

Table I. Certified statements used in this paper (Lean 4, commit `42f307b`) and their status in the literature.

Lean theorem (prefix `QSL.`)	statement	status
margolus_levitin	$\pi \le 2\langle E\rangle\tau$	classical [ML]
mandelstam_tamm_L1, _weak	$1 \le D_1\tau \le \Delta E\,\tau$	weak form of [MT]
zielinski_zych_all	$\pi^\alpha \le 2M_\alpha\tau^\alpha$, all $\alpha > 0$	stated in [ZZ]; key inequality unproved there
Qudit.arccos_le_entangling_time	$\arccos\lambda \le J\sum\tau_k$	corollary of [BJG,Rudnicki]
Qudit.pi_div_four_le_entangling_time_fin	Bell: $J\sum\tau_k \ge \pi/4$, any d	$d=2$: [Dur,VHC]
Qudit.entangling_time_tight	attained by $J\,Z\otimes X$	$d=2$: [Dur,VHC]
Qudit.pi_div_four_le_entangling_time_local, _shift_local	same for $(P\otimes Q)$ Bell (ECR)	corollary
TwoQubit.entangling_time_lower_bound	chord form, constant $\sqrt{2-\sqrt{2}}$	weaker earlier version
Qudit.arccos_le_entangling_time_shift	$\|H\| \to \|H - c\mathbb{1}\|$	corollary; $d=2$: [Dur]
Qudit.pi_div_eight_le_heisenberg_time	Heisenberg Bell: $T \ge \pi/8$	$d=2$, no ancilla: [Dur]

Gate.fsAngle_le_time_halfWidth	angle $\le (\Omega_{\max}/2)\,T$, piecewise $H(t)$	standard (shifted MT)
FixedAxis.mandelstam_tamm_integral_eq_solution	$\int \Delta E = \arccos\lvert\langle\psi_0\vert\psi_T\rangle\rvert$, fixed axis	equality case of [MT]
Qudit.pi_div_four_sub_le_entangling_time_approx	Bell to fidelity $1-\varepsilon$: $J\sum\tau_k \ge \pi/4 - \arcsin\sqrt{\varepsilon}$	robust corollary; formalized here
Gate.two_arccos_sqrt_le_time_approx	transfer to fidelity $1-\varepsilon$: $\Omega_{\max}T \ge 2\arccos\sqrt{\varepsilon}$	two-level; tight
Exchange.exchange_norm_ge, _le, ratio_tendsto_one	$\lambda_d^2/2 \le \kappa_d \le 2(d-1)$; $2\kappa_d/\lambda_d^2 \to 1$	not found in the sources read
Exchange.refined_cz_lemma	$\frac{1}{4}\mathrm{Tr}[(P_A\otimes P_B)H_c^2] \le \frac{97}{32}J^2$, rank-2 codes, $d=3$	lemma of the refined CZ bound

Colour: red = no / fails.

p. 4 in the PDF | ⬇ CSV | cited in §1 Introduction | cited in §2 Certified Bounds | cited in §9 What the Certificate Does and Does Not Guarantee | cited in §11 Conclusion

Table II. Single-qubit gap, IBM snapshots (`qiskit-ibm-runtime` 0.51.0; 41 backends, 1646 qubits). $\tau_{\mathrm{QSL}} = \pi/\texttt{omegad}$ for X (full drive amplitude, $|D| = 1$).

quantity	median	percentiles
$\tau_X/\tau_{\mathrm{QSL}}$ (5th–95th)	11.30	[7.19, 15.54]
$\tau_{\sqrt{X}}/\tau_{\mathrm{QSL}}(\sqrt{X})$	22.61	—
$\tau_X/(2\pi/\lvert\delta\rvert)$ (5th–95th)	15.45	[10.90, 18.70]
$\Omega_{\mathrm{peak}}/\lvert\delta\rvert$ (750 qubits, 10th–90th)	0.0907	[0.0607, 0.0992]

Colour: blue shade = the number's size within its column.

p. 4 in the PDF | ⬇ CSV | cited in §4.1 The gap

Table III. Two-qubit gap (1758 pairs, 43 backends; 1019 ECR, 739 CX). Qubit subspace, $\|H_{\mathrm{int}}\| = j_q$.

bound	source	median ratio
CNOT-class gate, $\pi/(2j_q)$	[Howard,VHC] (not in Lean)	4.22
Bell preparation, $\sqrt{2-\sqrt{2}}/j_q$	Lean, chord form	8.7
Bell preparation, $\pi/(4j_q)$	Lean, sharp	2× row 1

p. 6 in the PDF | ⬇ CSV | cited in §5.3 Gaps

Table IV. Median gaps by year (56 usable snapshots, 41 backends). Two-qubit ratio against $\pi/(2j_q)$.

	2021	2022	2023	2024	2025
$\tau_X/\tau_{\mathrm{QSL}}$	11.28	11.32	11.21	11.10	11.87
$\tau_{2q}/(\pi/2j_q)$	2.78	2.64	4.32	3.89	4.26
snapshots	27	4	2	16	7

Colour: blue shade = the number's size within its column.

p. 7 in the PDF | ⬇ CSV | cited in §3.4 Exclusions and falsifier | cited in §6 Time Series 2021--2025

Table V. Shortest X-gate time found by optimal control (an upper bound on the minimum), 4-level transmon, error $< 10^{-4}$ with leakage, units $|\delta| = 1$ (`grape_1q.py`).

$r = \Omega_{\max}/\lvert\delta\rvert$	T_{found}	$T_{\mathrm{found}}/(\pi/\Omega_{\max})$
0.1	33.0	1.05
0.2	16.5	1.05
0.3	12.0	1.15
0.5	8.2	1.30
0.8	7.1	—
1.0	6.9	—
1.5	7.3	—

Colour: blue shade = the number's size within its column.

p. 8 in the PDF | ⬇ CSV | cited in §4.1 The gap | cited in §4.3 What sets the duration | cited in §7.1 Single qubit | cited in §9 What the Certificate Does and Does Not Guarantee

Table VI. CZ on the qubit subspace of two exchange-coupled 3-level transmons: best error $1 - F$ against gate time T and the bound on each local control $\|A_k\|, \|B_k\| \le a_{\max}$ (`grape_cz_torch.py`, `verificar_cz.py`). Qubit-only limit $\pi/(2J) = 1.571/J$; lower bounds $\pi/(8J) = 0.393/J$ (certified) and $0.4511/J$ (refined, Section VII); explicit construction $\pi/(6J) = 0.524/J$. Each cell is the best of up to 12 (no cap) or 8 (capped) random starts; differences below 10^{-8} between columns are optimizer noise.

$T\,J$	none	$40J$	$20J$	$10J$

0.40	$1.6 \cdot 10^{-2}$			
0.45	$5.0 \cdot 10^{-3}$	$5.3 \cdot 10^{-3}$	$5.5 \cdot 10^{-3}$	
0.50	$3.7 \cdot 10^{-4}$	$4.2 \cdot 10^{-4}$	$4.5 \cdot 10^{-4}$	$1.3 \cdot 10^{-3}$
0.55	$7 \cdot 10^{-12}$	$8 \cdot 10^{-13}$	$2 \cdot 10^{-11}$	$2 \cdot 10^{-8}$
0.60	$3 \cdot 10^{-10}$			$1 \cdot 10^{-9}$

Colour: blue shade = the number's size within its column.

p. 9 in the PDF ⬇ CSV

cited in §9 What the Certificate Does and Does Not Guarantee

Table VII. IBM chip-to-chip transfers [Sundaresan] against the single-mode optimum $\pi/(\sqrt{2}\,g)$, for $g/2\pi = 8$ and 9 MHz (44.2 and 39.3 ns). Error $= 1 - F$ as reported. Coherence at T_g: IBM's error model (their Eqs. (1)–(2), occupations $\frac{1}{2}$ qubits, $\frac{1}{2}$ cable; our reproduction). Coherence at $T_{\min}$: Lindblad master equation of the optimal protocol with the same coherence times. v:gapsInteractive version for your own link.

Link	T_g (ns)	$T_g/T_{\min}$	error	coh. at T_g	coh. at $T_{\min}$
Q240–Q247	312	7.06–7.94	0.71%	0.63%	0.13–0.15%
Q241–Q246	296	6.70–7.53	1.13%	0.87%	0.23–0.26%
Q242–Q245	256	5.79–6.52	1.01%	0.61%	0.16–0.18%
Q243–Q244	312	7.06–7.94	1.14%	0.94%	0.22–0.24%

Colour: blue shade = the number's size within its column.

p. 10 in the PDF ⬇ CSV cited in §8 Application: A Chip-to-Chip State Transfer

References

1

L. Mandelstam and I. Tamm, "The uncertainty relation between energy and time in non-relativistic quantum mechanics," *J. Phys. USSR*, vol. 9, pp. 249–254, 1945. 🔎 find it

2

N. Margolus and L. B. Levitin, "The maximum speed of dynamical evolution," *Physica D*, vol. 120, pp. 188–195, 1998. 🔎 find it

3

B. Zieliński and M. Zych, "A generalization of Margolus–Levitin bound," *Phys. Rev. A*, vol. 74, 034301, 2006. arXiv:quant-ph/0603076.

4

W. Dür, G. Vidal, J. I. Cirac, N. Linden, and S. Popescu, “Entanglement capabilities of nonlocal Hamiltonians,” *Phys. Rev. Lett.*, vol. 87, 137901, 2001. arXiv:quant-ph/0006034.

5

G. Vidal, K. Hammerer, and J. I. Cirac, “Interaction cost of nonlocal gates,” *Phys. Rev. Lett.*, vol. 88, 237902, 2002. arXiv:quant-ph/0112168.

6

K. Hammerer, G. Vidal, and J. I. Cirac, “Characterization of non-local gates,” *Phys. Rev. A*, vol. 66, 062321, 2002. arXiv:quant-ph/0205100.

7

B. Basyildiz, C. Jameson, and Z. Gong, “Speed limits of two-qubit gates with qudits,” arXiv:2312.09218, 2023.

8

B. Basyildiz, Z. Gong, and S. Ashhab, “Speed limits of two-qutrit gates,” arXiv:2510.07742, 2025.

9

J. Howard, A. Lidiak, C. Jameson, B. Basyildiz, K. Clark, T. Zhao, M. Bal, J. Long, D. P. Pappas, M. Singh, and Z. Gong, “Implementing two-qubit gates at the quantum speed limit,” *Phys. Rev. Research*, vol. 5, 043194, 2023. arXiv:2206.07716.

10

Ł. Rudnicki, “Quantum speed limit helps interpret geometric measure of entanglement,” *Phys. Rev. A*, vol. 104, 032417, 2021. arXiv:2107.11877.

11

S. Bravyi, “Upper bounds on entangling rates of bipartite Hamiltonians,” *Phys. Rev. A*, vol. 76, 052319, 2007. arXiv:0704.0964.

12

N. Ishida and Y. Hasegawa, “Energy inference of black-box quantum computers using quantum speed limit,” arXiv:2512.15472, 2025.

13

D. Basilewitsch, C. Dlaska, and W. Lechner, “Comparing planar quantum computing platforms at the quantum speed limit,” *Phys. Rev. Research*, vol. 6, 023026, 2024. arXiv:2304.01756.

14

Y. Wu, J. Yuan, C. Zhang, Z. Zhu, J. Deng, X. Zhang, P. Zhang, Q. Guo, Z. Wang, J. Huang, C. Song, H. Li, D.-W. Wang, H. Wang, and G. S. Agarwal, “Testing the unified bounds of quantum speed limit,” *Phys. Rev. A*, vol. 110, 042215, 2024. arXiv:2403.03579.

15

M. Werninghaus, D. J. Egger, F. Roy, S. Machnes, F. K. Wilhelm, and S. Filipp, “Leakage reduction in fast superconducting qubit gates via optimal control,” *npj Quantum Inf.*, vol. 7, 14, 2021. arXiv:2003.05952.

16

S. Ashhab, F. Yoshihara, T. Fuse, N. Yamamoto, A. Lupascu, and K. Semba, “Speed limits for two-qubit gates with weakly anharmonic qubits,” *Phys. Rev. A*, vol. 105, 042614, 2022. 🔎 find it

17

P. Wocjan, M. Rötteler, D. Janzing, and T. Beth, “Simulating Hamiltonians in quantum networks: Efficient schemes and complexity bounds,” *Phys. Rev. A*, vol. 65, 042309, 2002. arXiv:quant-ph/0109088.

18

E. Magesan and J. M. Gambetta, “Effective Hamiltonian models of the cross-resonance gate,” *Phys. Rev. A*, vol. 101, 052308, 2020. arXiv:1804.04073.

19

Qiskit IBM Runtime, version 0.51.0, `fake_provider` backends, Apache 2.0. https://github.com/Qiskit/qiskit-ibm-runtime.

20

Qiskit Terra, version 0.24.1, `fake_provider` backends, Apache 2.0. https://github.com/Qiskit/qiskit (package `qiskit-terra` on PyPI).

21

N. M. Sundaresan *et al.*, “Meter-scale quantum interconnects between large quantum processors,” arXiv:2610.29986 (2026).

22

U. Boscain, T. Chambrion, and G. Charlot, “Nonisotropic 3-level quantum systems: complete solutions for minimum time and minimum energy,” *Discrete Contin. Dyn. Syst. Ser. B*, vol. 5, pp. 957–990, 2005; arXiv:quant-ph/0409022.

23

C. Marín, “Quantum Speed Limits: the Zieliński–Zych Bound and Time–Occupation Bounds for Qubit–Bus–Qubit State Transfer,” Isabelle/HOL, submitted to the Archive of Formal Proofs (October 2026); sources in the companion package. 🔎 find it

24

K. Heya *et al.*, “Randomized benchmarking of a remote CNOT gate via a meter-scale microwave link,” *Phys. Rev. Lett.*, vol. 135, art. 200801, 2025; arXiv:2502.15034.

25

C. Marín, “Formalizations” (Lean 4), https://github.com/karlesmarin/formalizations, commit `42f307b`, 2026. Companion text: C. Marín, “The clock that never ticks: a

machine-checked path from the dilogarithm to quantum speed limits in Lean 4,” Zenodo, 2026, doi:10.5281/zenodo.20675271.